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LEI et al.(10) **Pub. No.: US 2025/0283614 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **WINDOW AIR CONDITIONER**(30) **Foreign Application Priority Data**(71) Applicants: **WUHU MATY**
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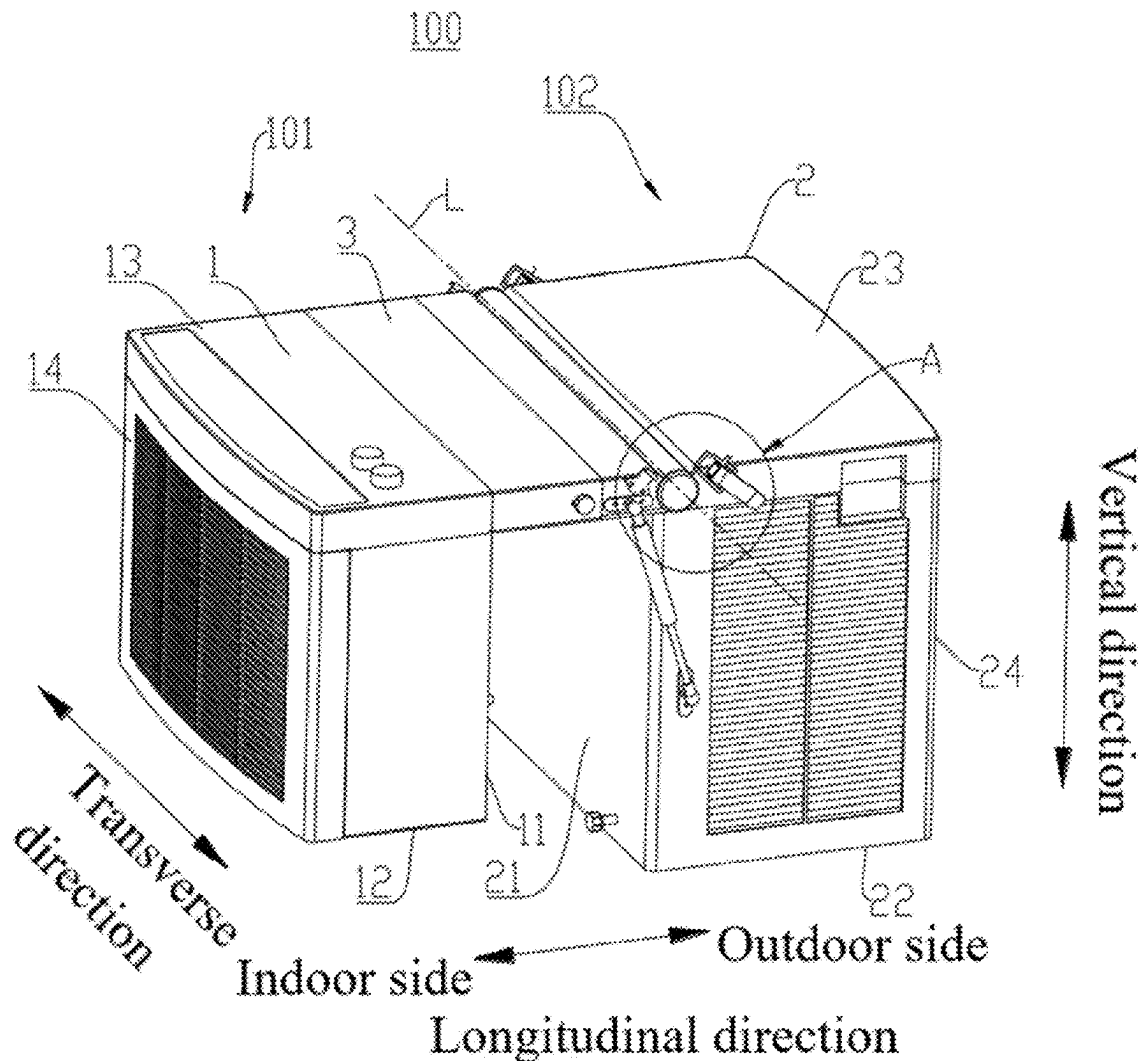
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CPC **F24F 1/031** (2019.02)(72) Inventors: **Zhisheng LEI**, Wuhu (CN); **Chengli**
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Kangwen ZHANG, Wuhu (CN);
Wenjun SHEN, Wuhu (CN)(57) **ABSTRACT**

A window air conditioner includes an indoor unit component and an outdoor unit component. The indoor unit component includes an indoor unit body. The outdoor unit component includes an outdoor unit body spaced apart from the indoor unit body in a longitudinal direction. The indoor unit component is rotatably connected to the outdoor unit component, to enable the outdoor unit component and the indoor unit component to rotate relative to each other about a rotation axis extending in a transverse direction.

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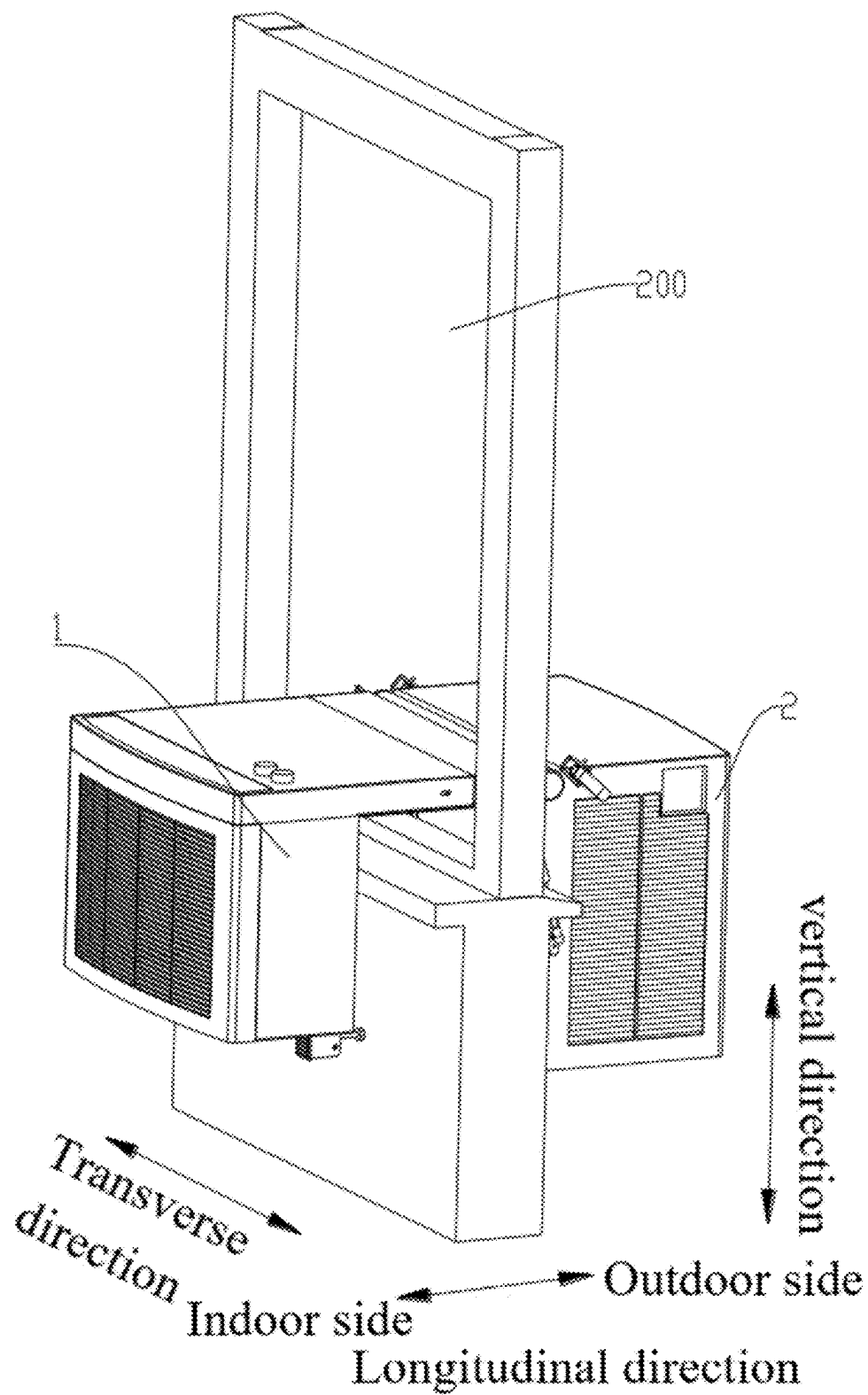


FIG. 2

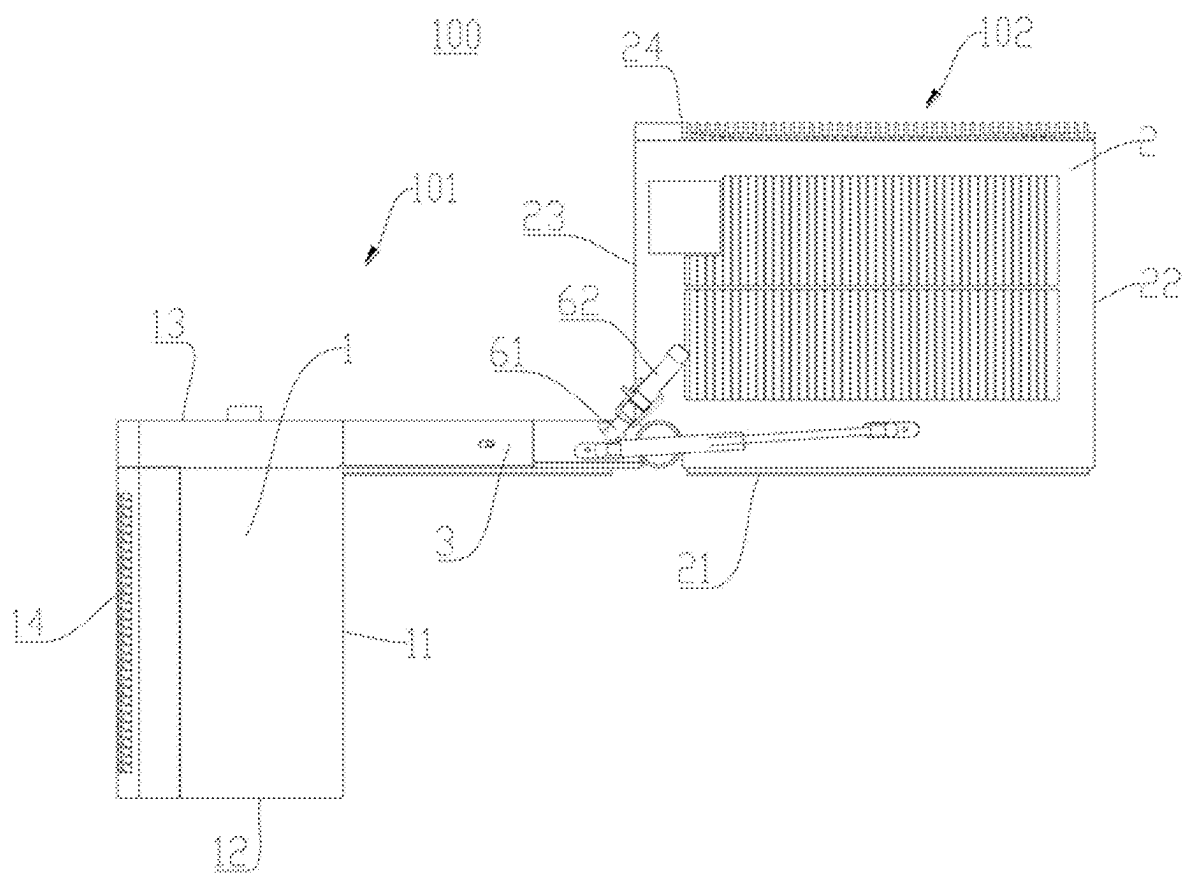


FIG. 3

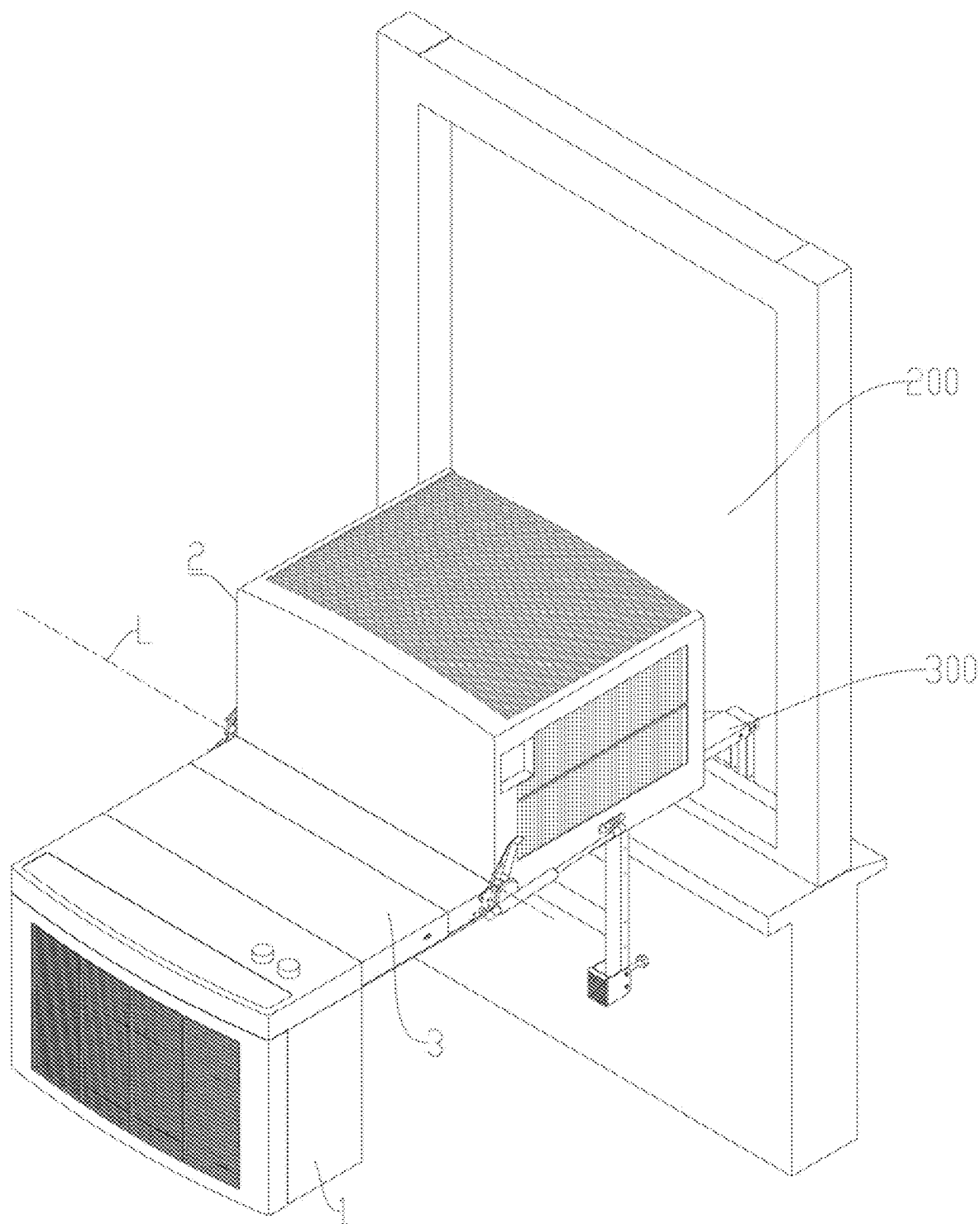
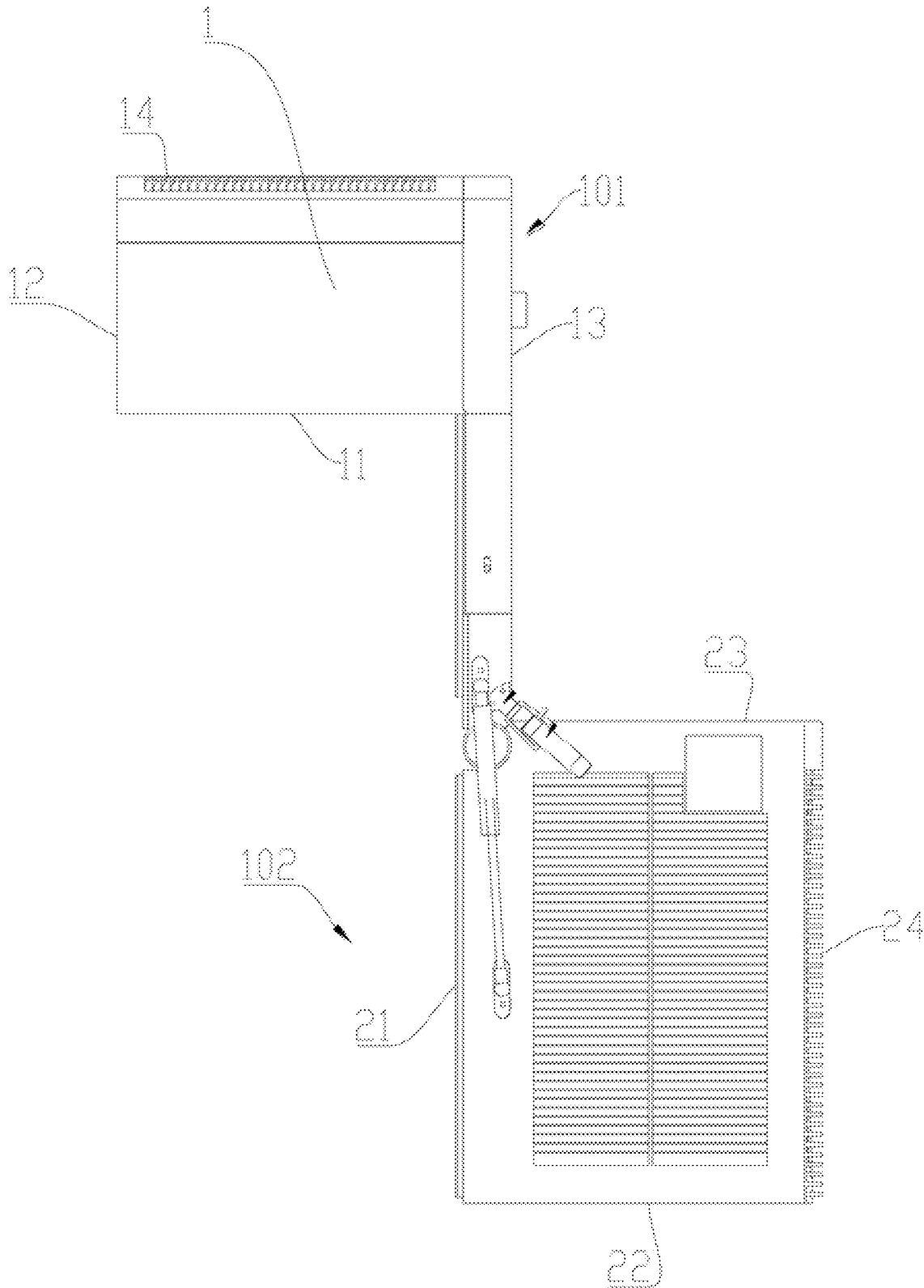


FIG. 4



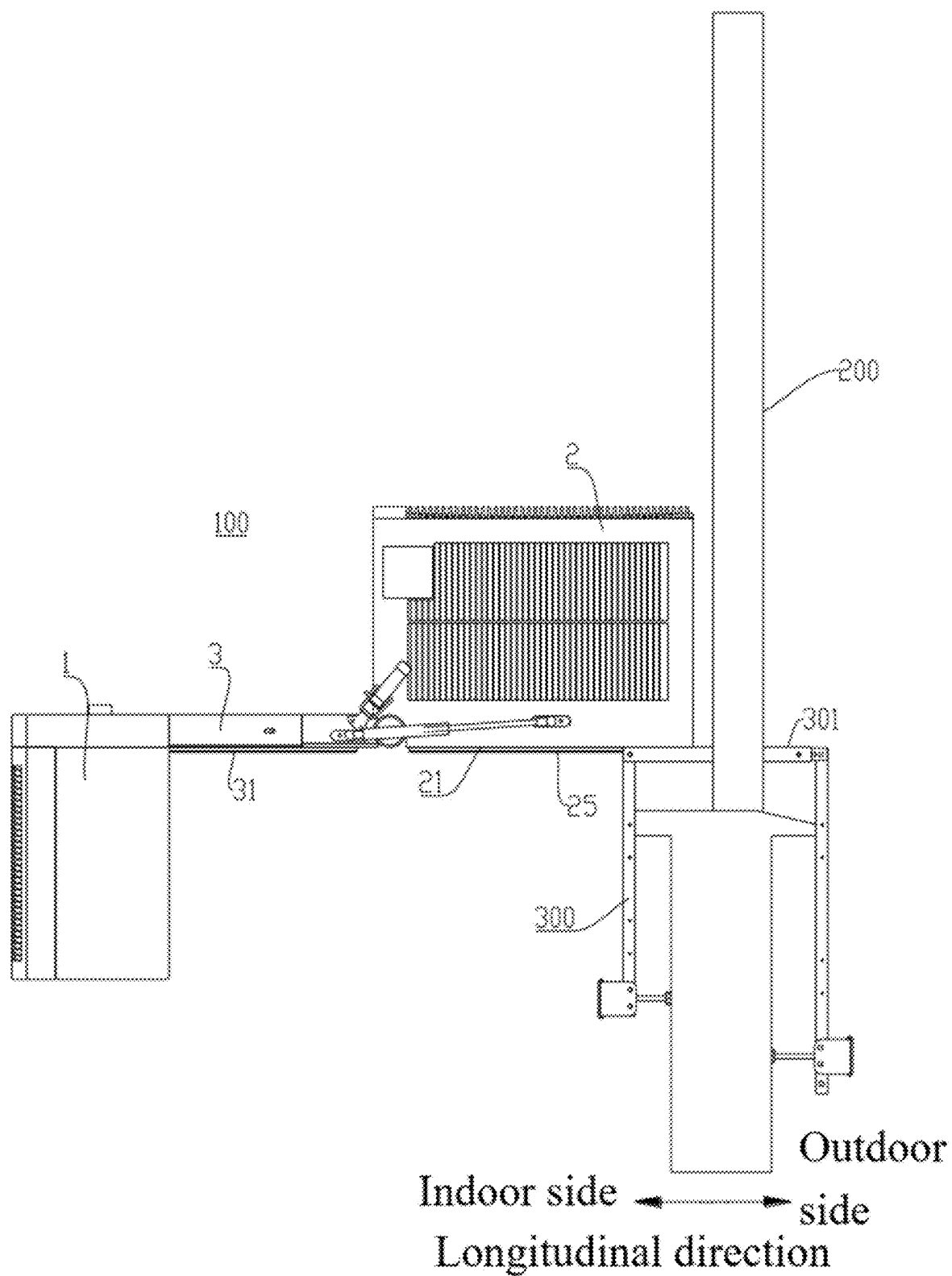
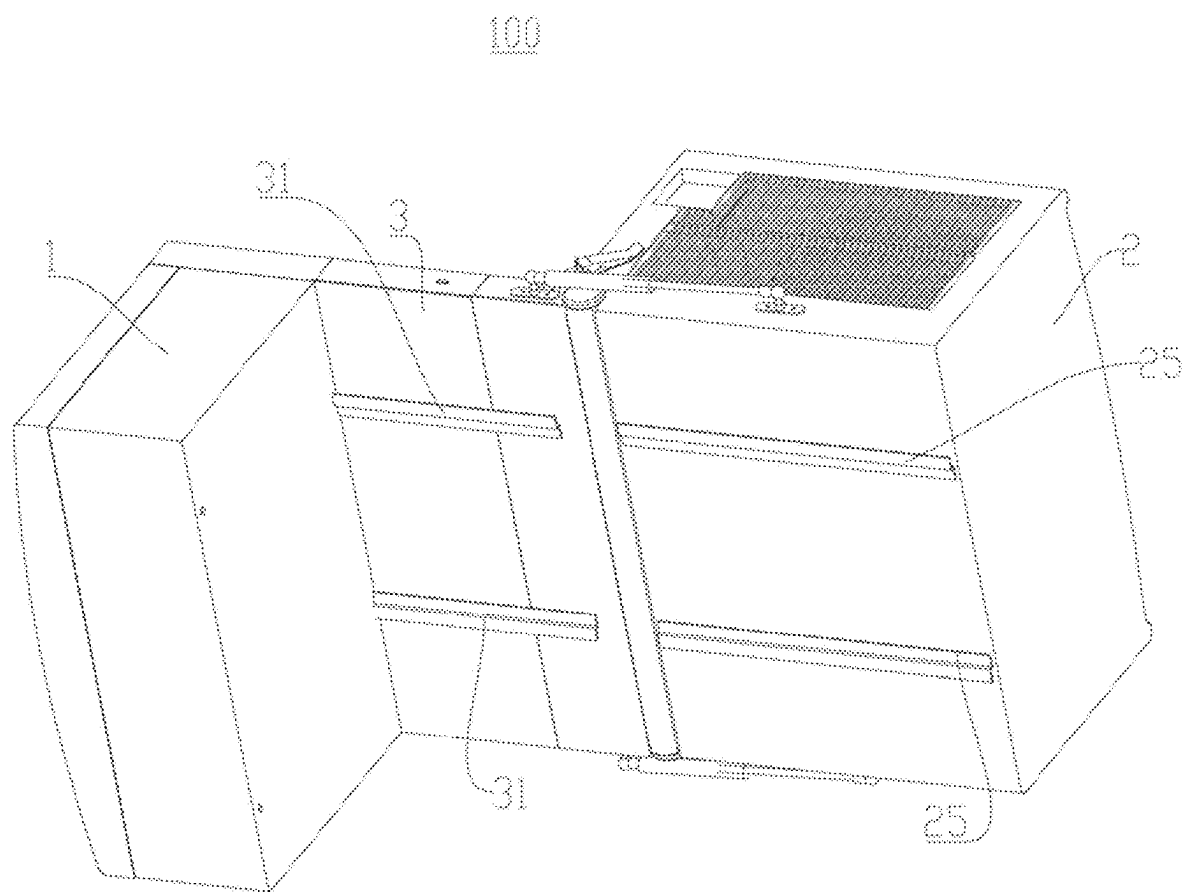


FIG. 6



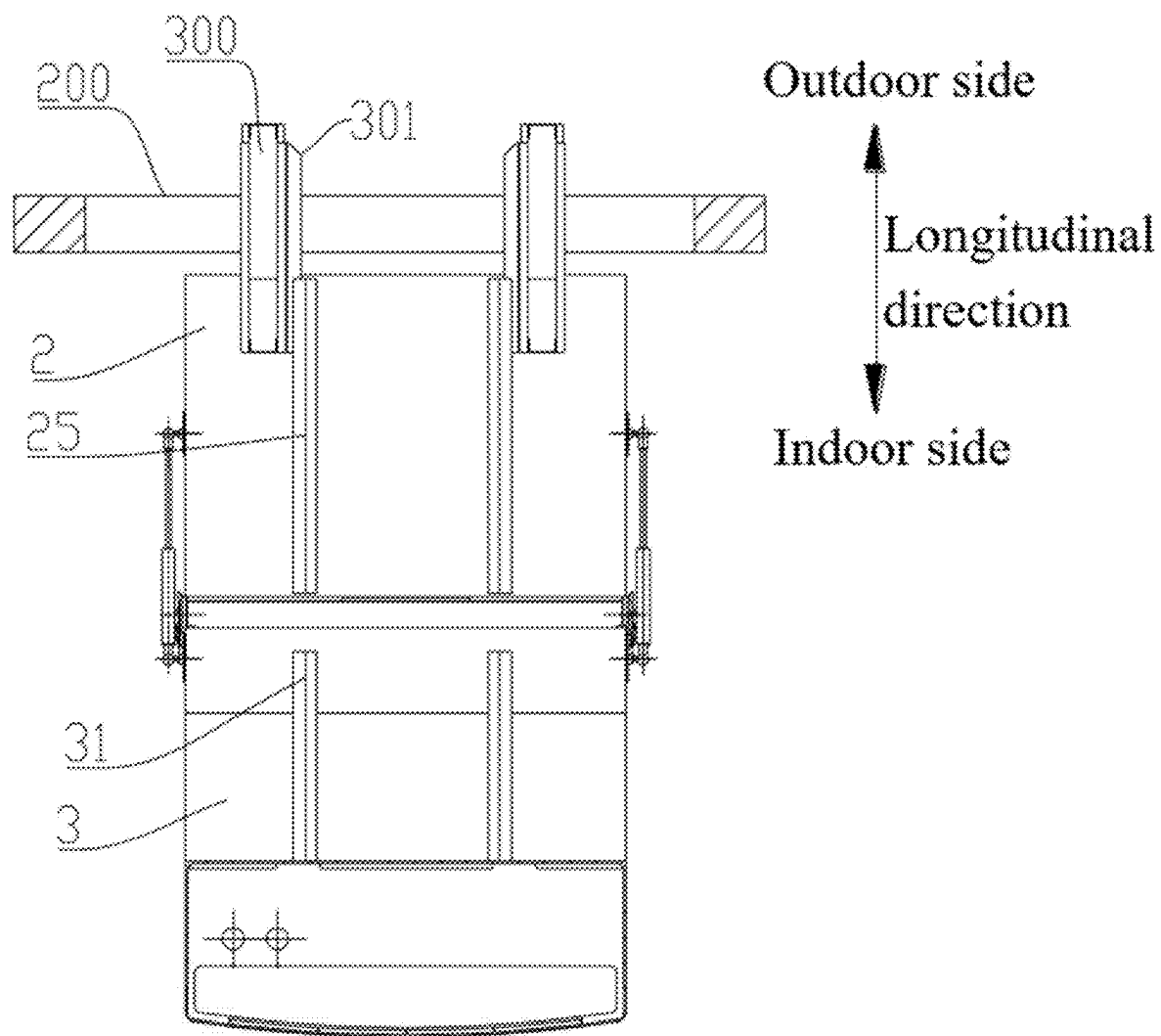


FIG. 8

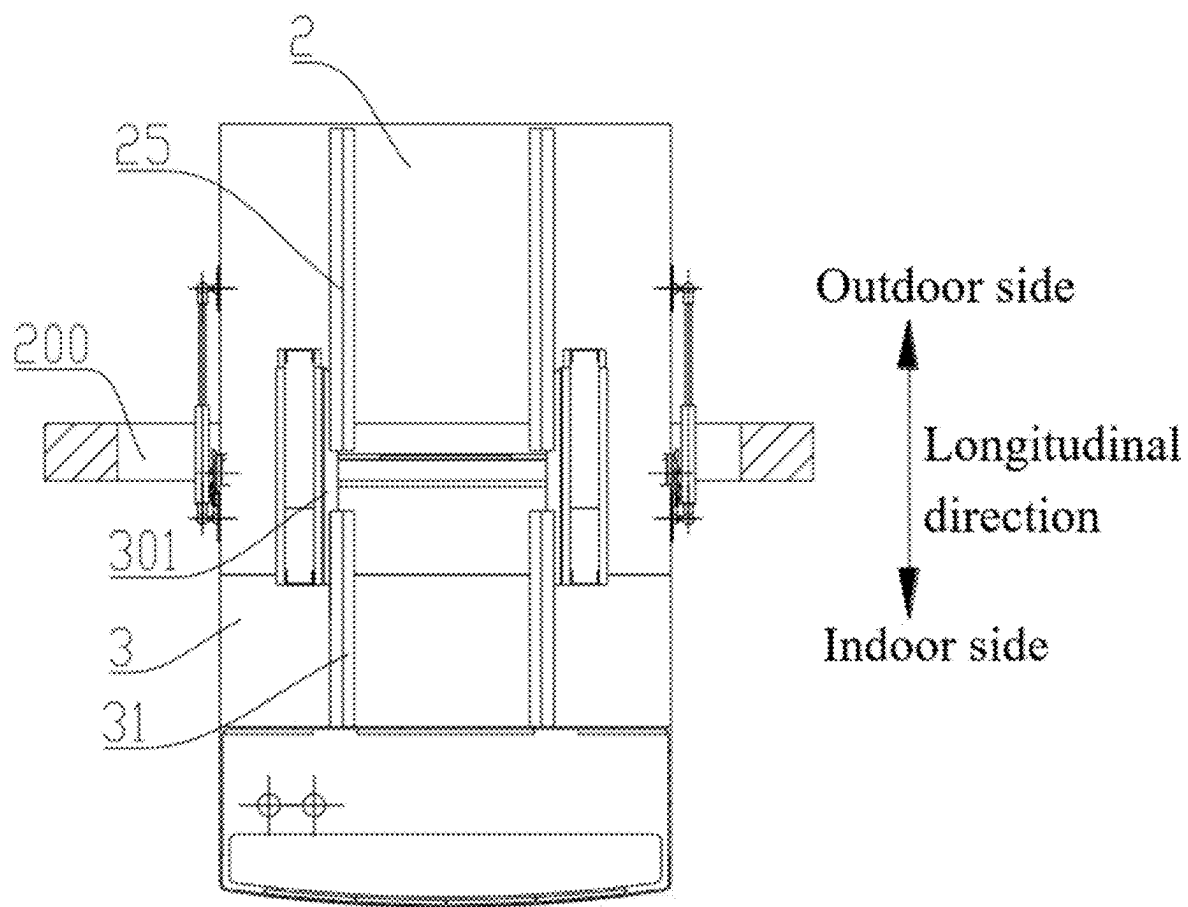


FIG. 9

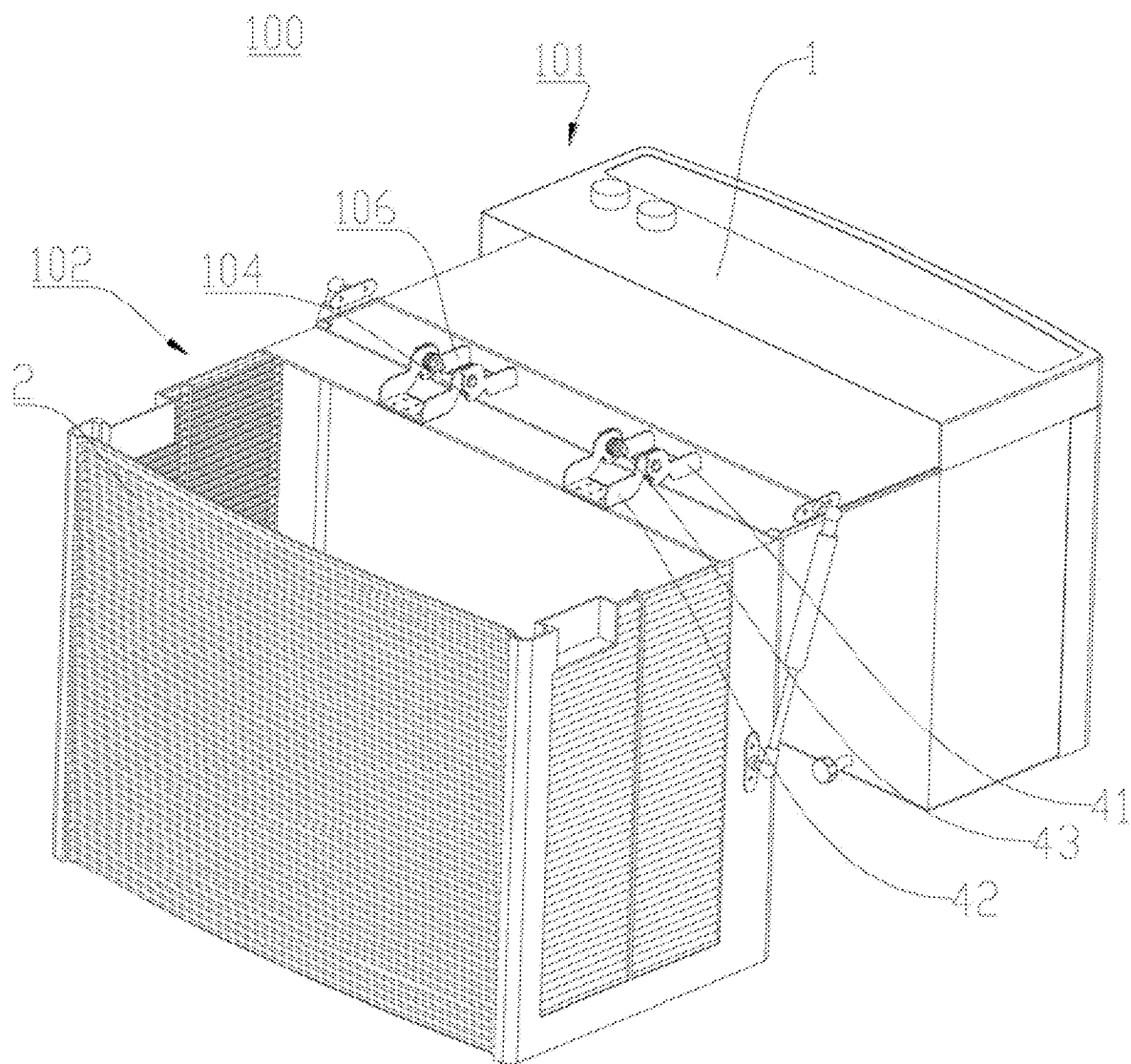


FIG. 10

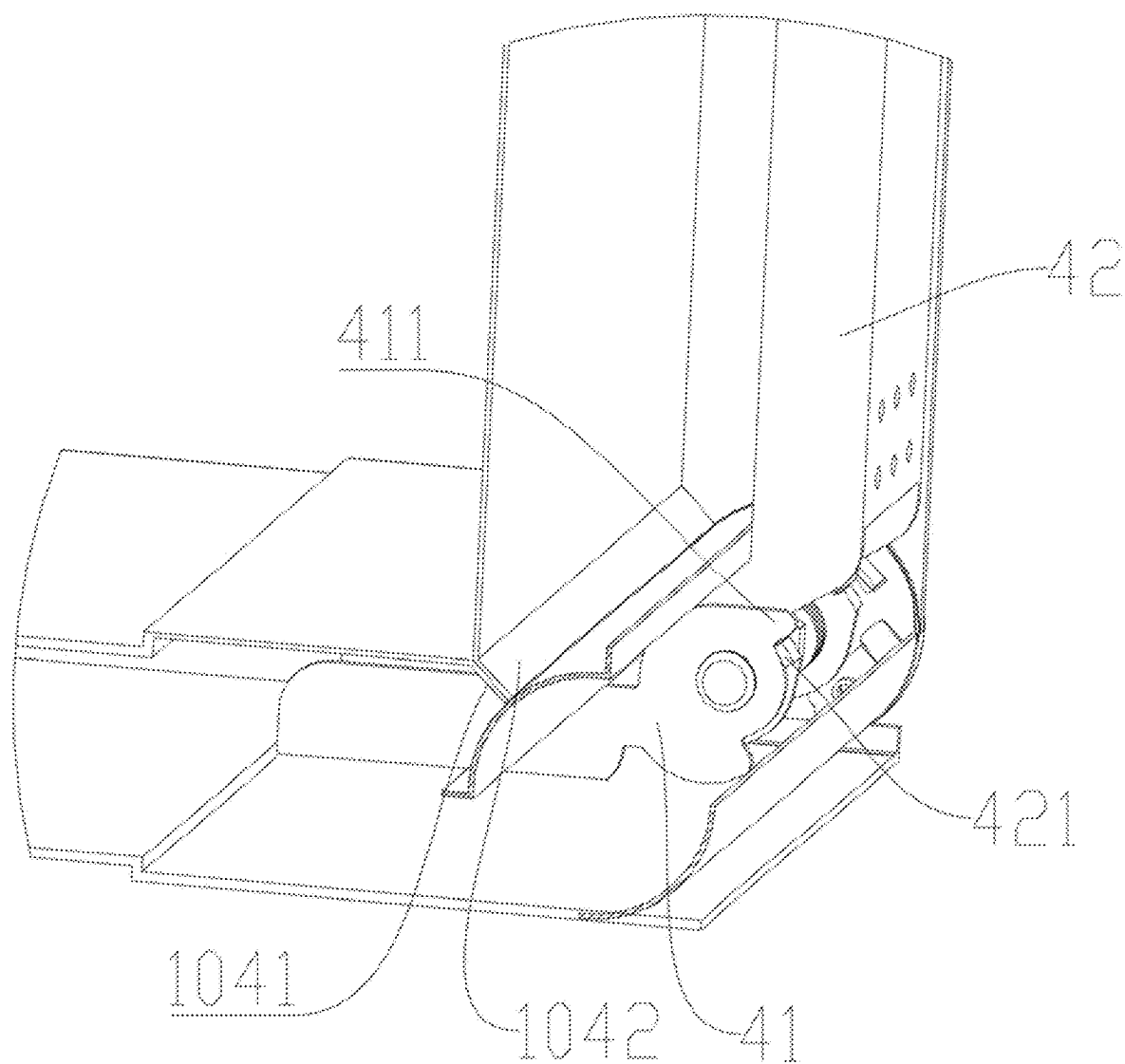


FIG. 11

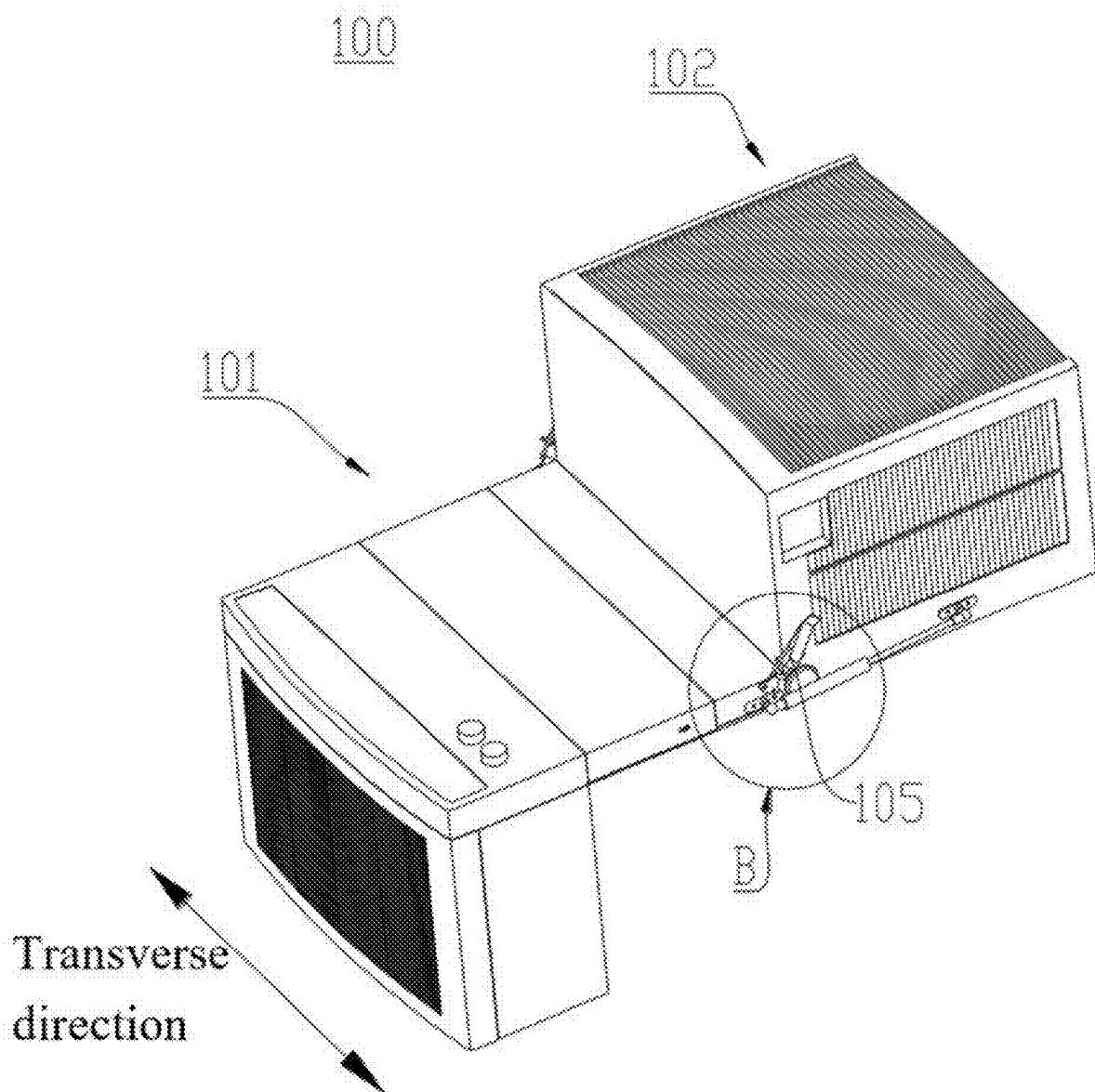


FIG. 12

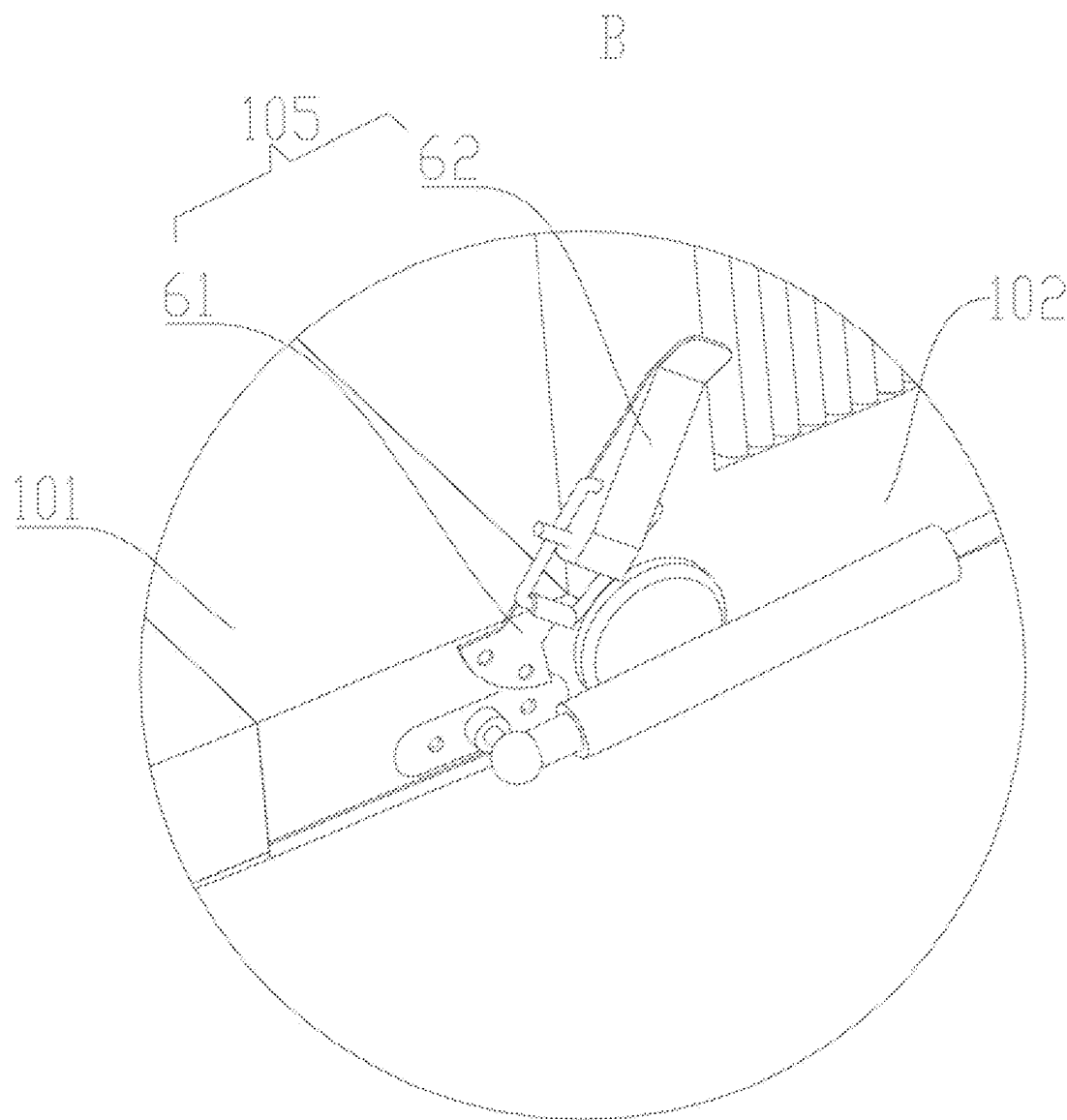


FIG. 13

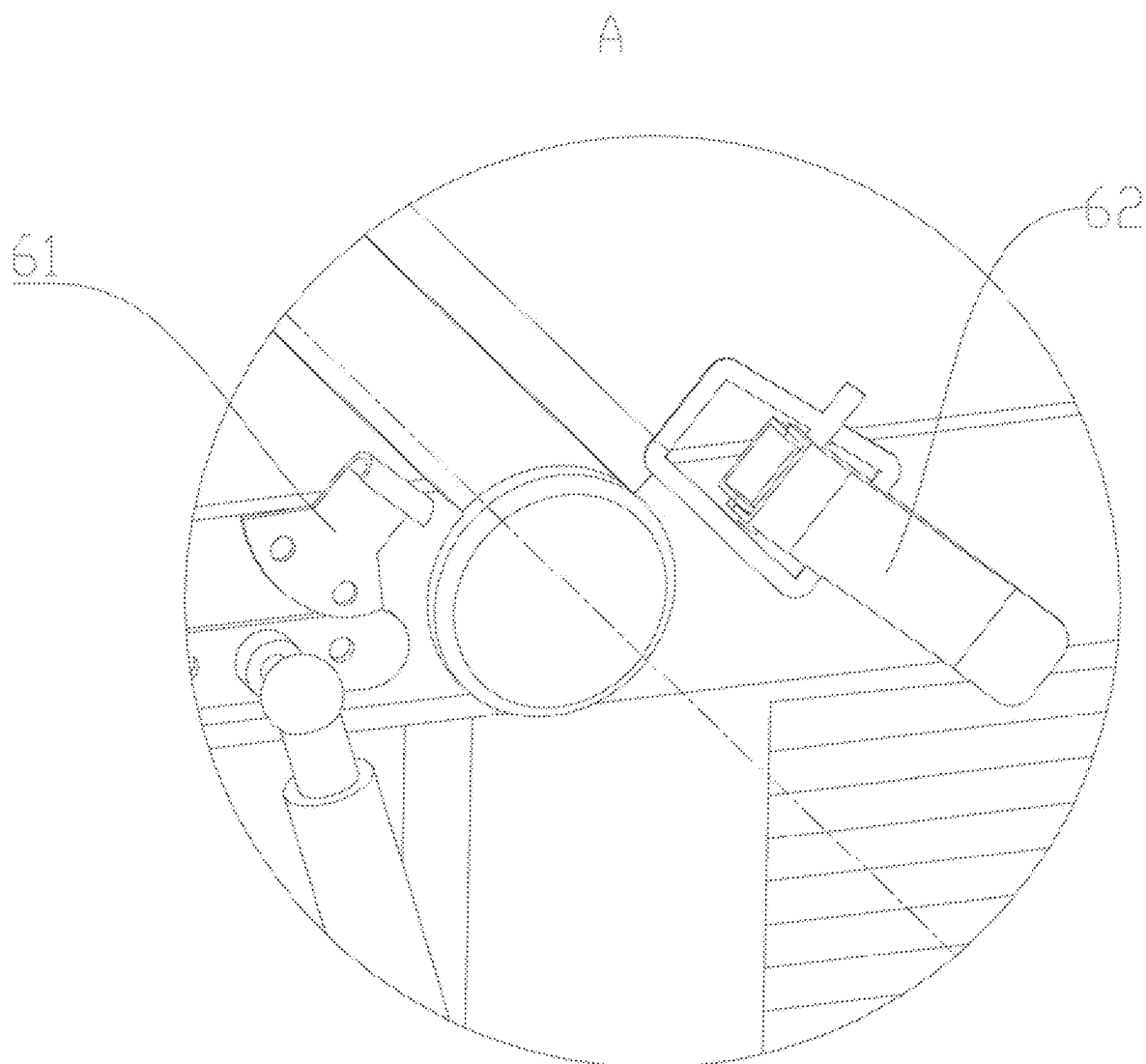


FIG. 14

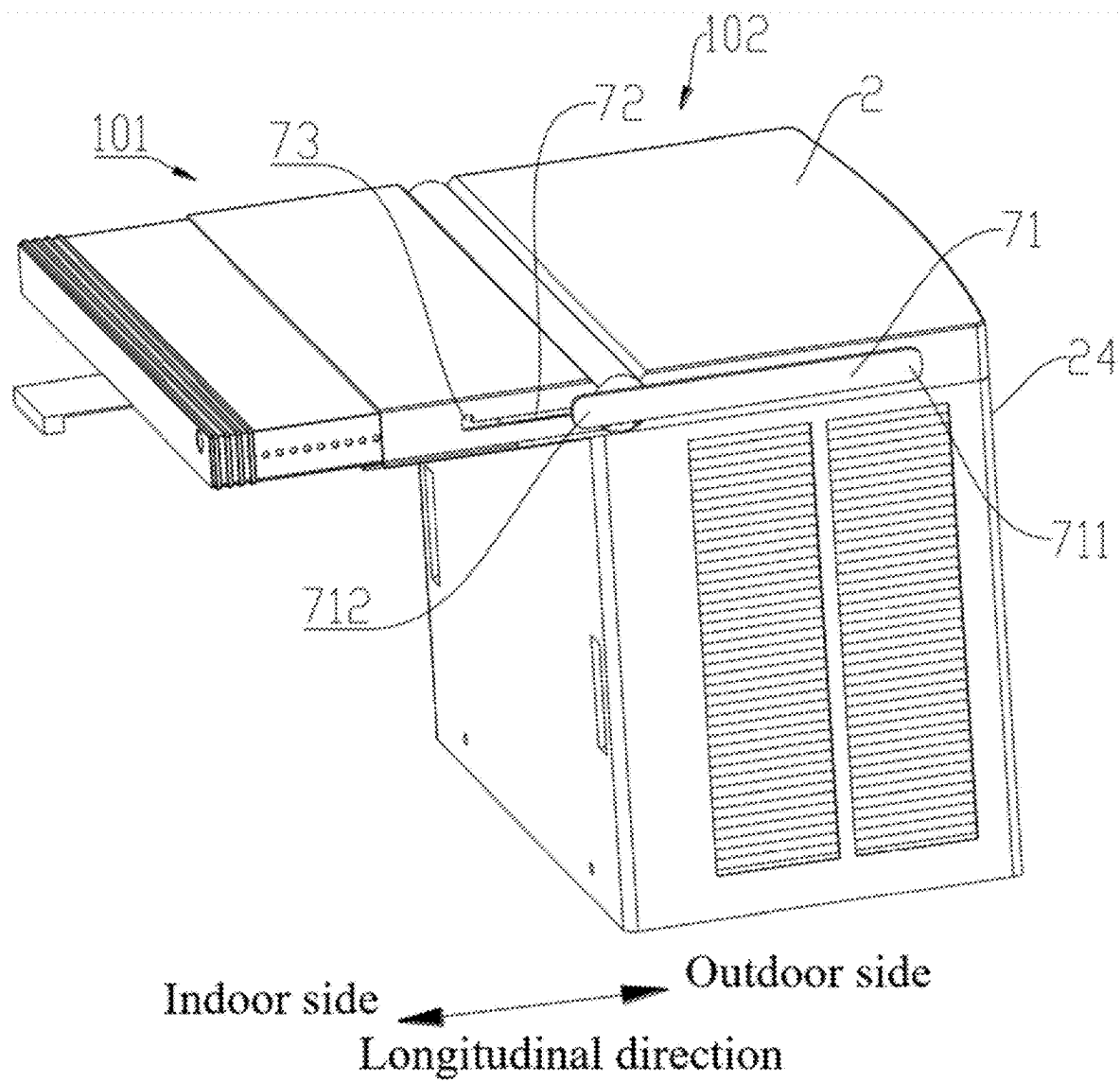


FIG. 15

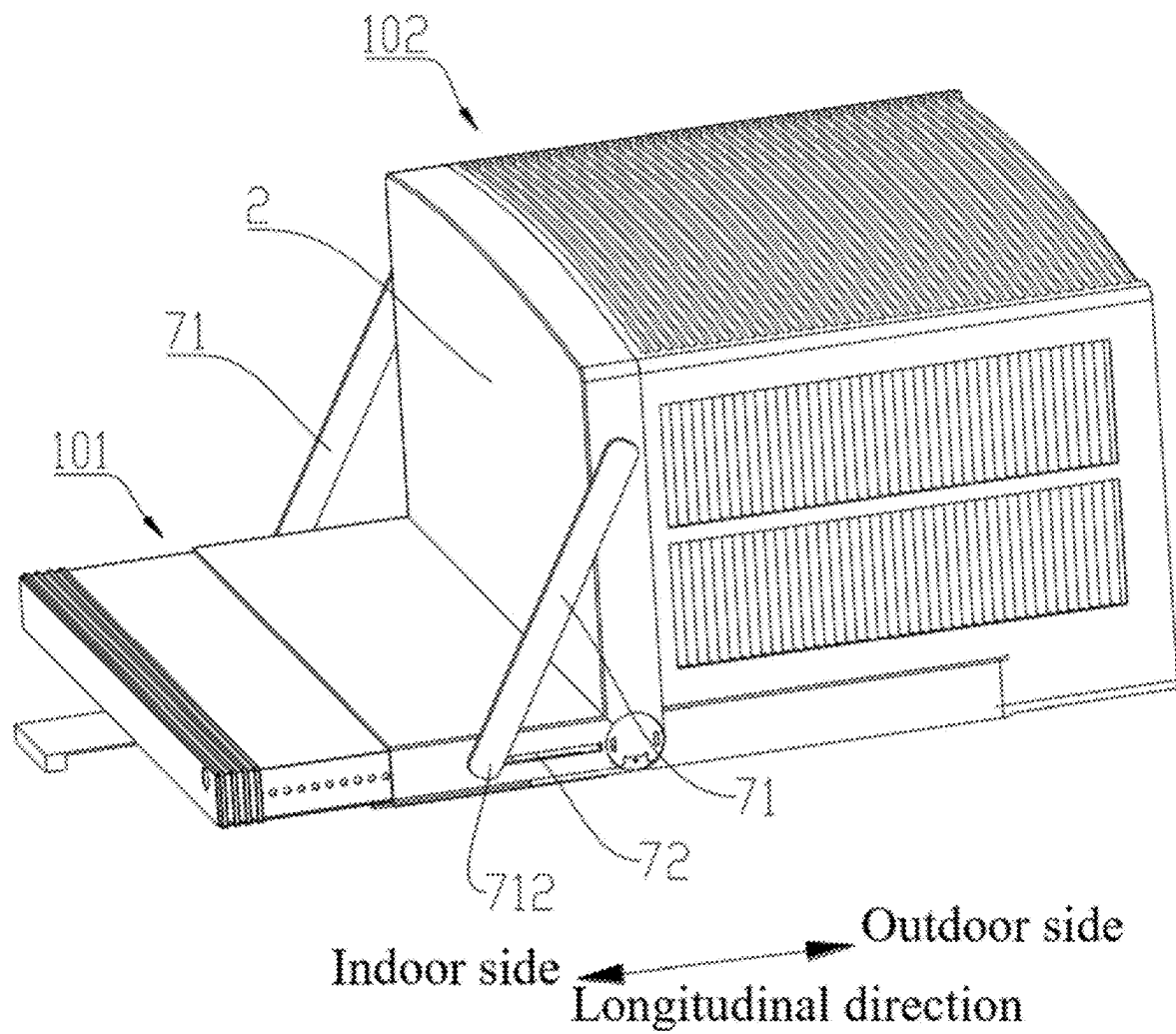


FIG. 16

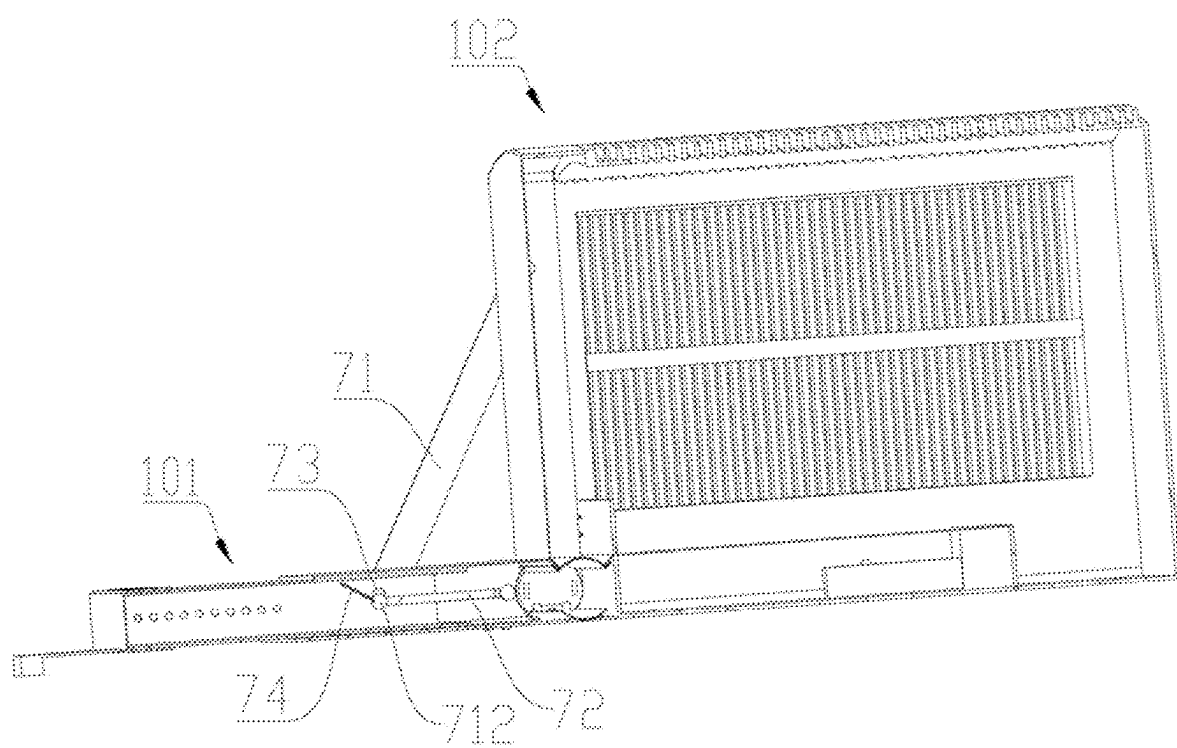


FIG. 17

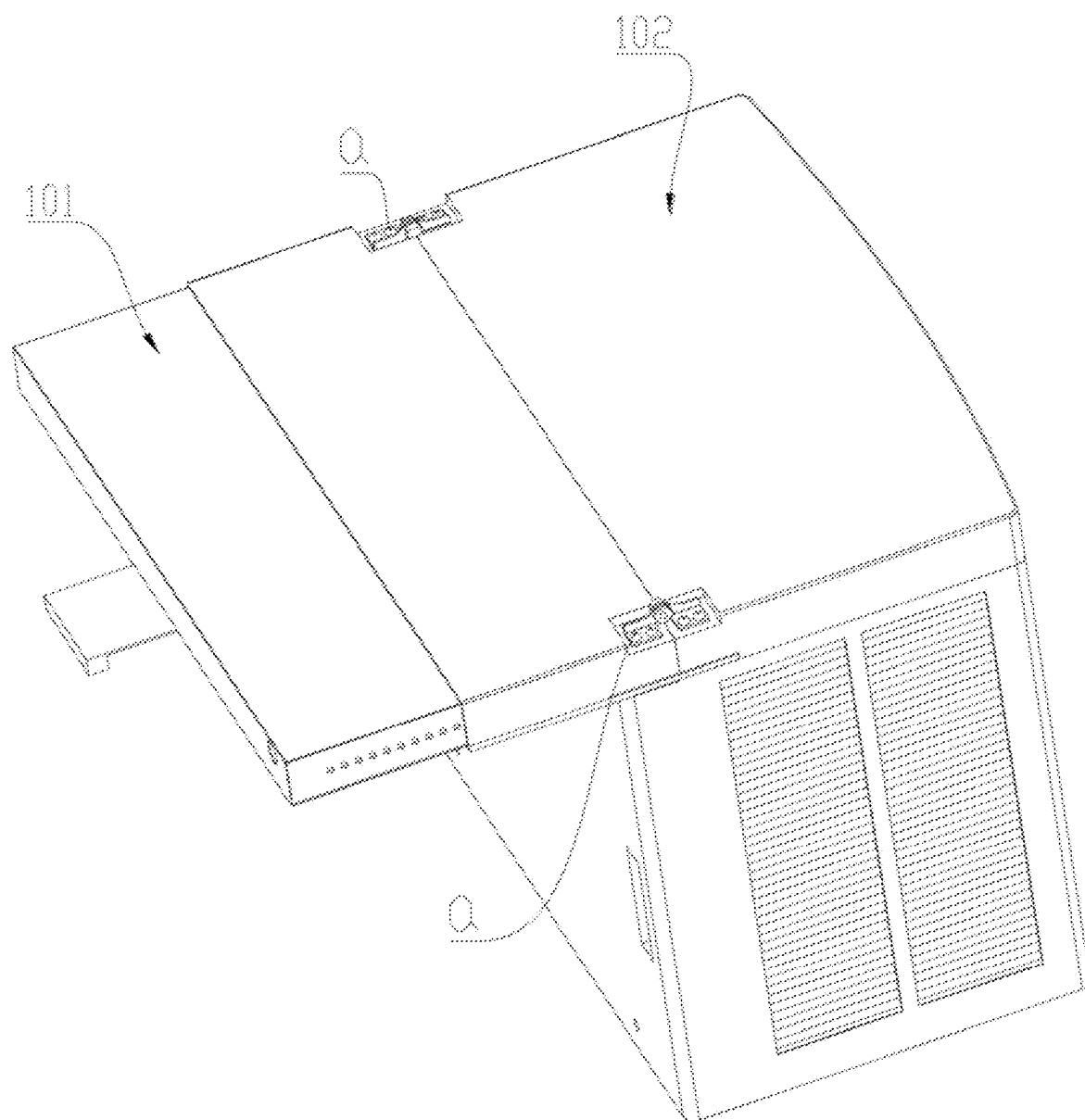


FIG. 18

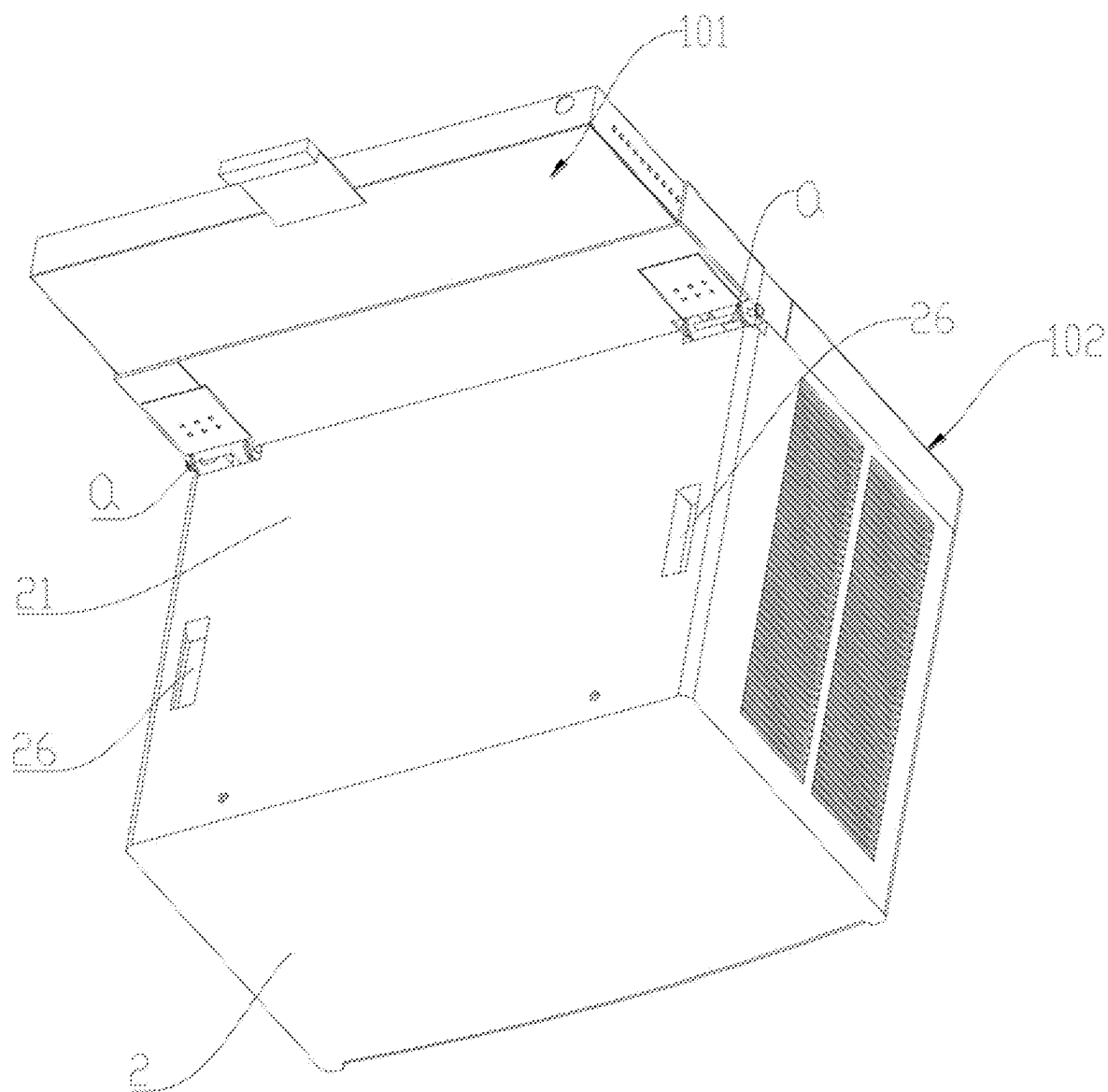


FIG. 19

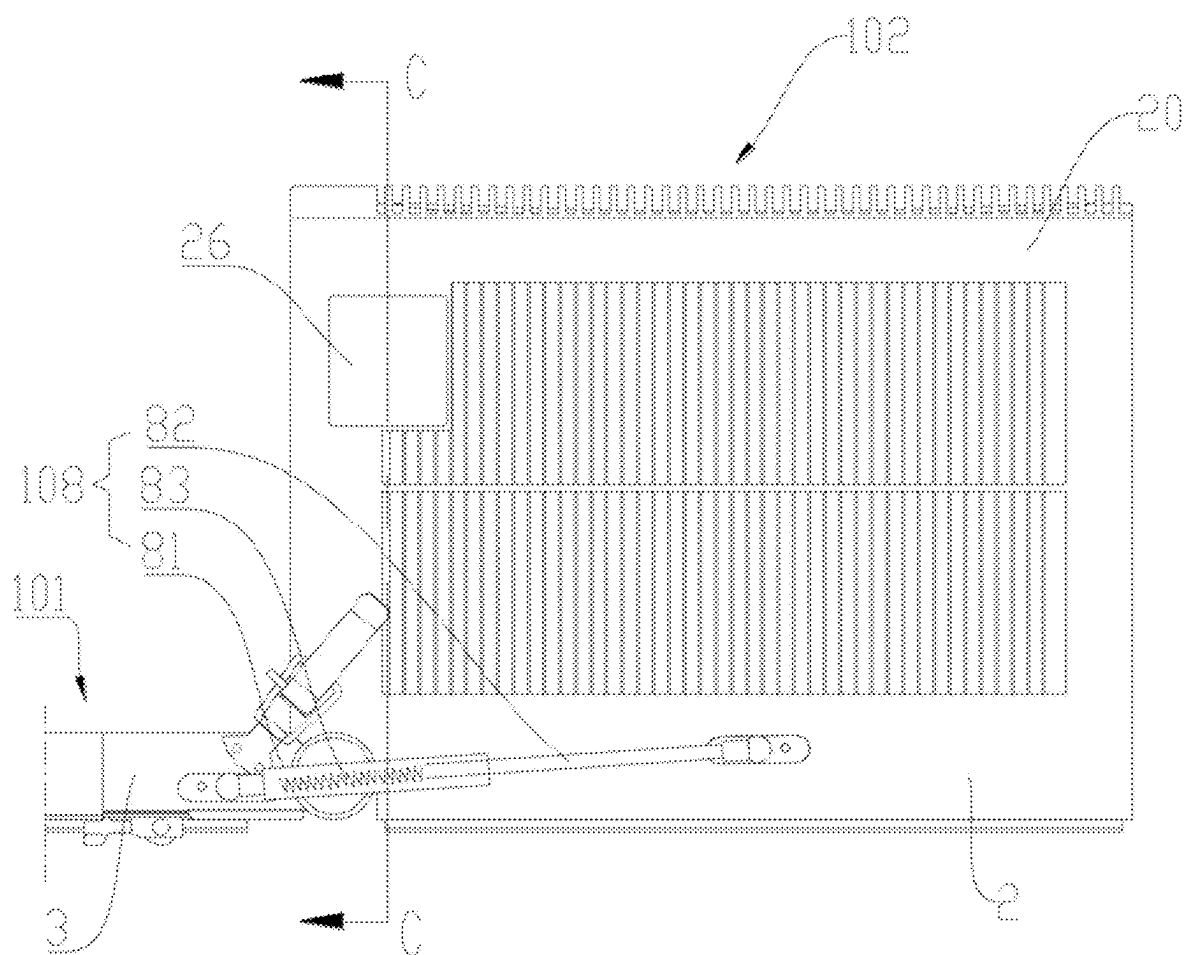


FIG. 20

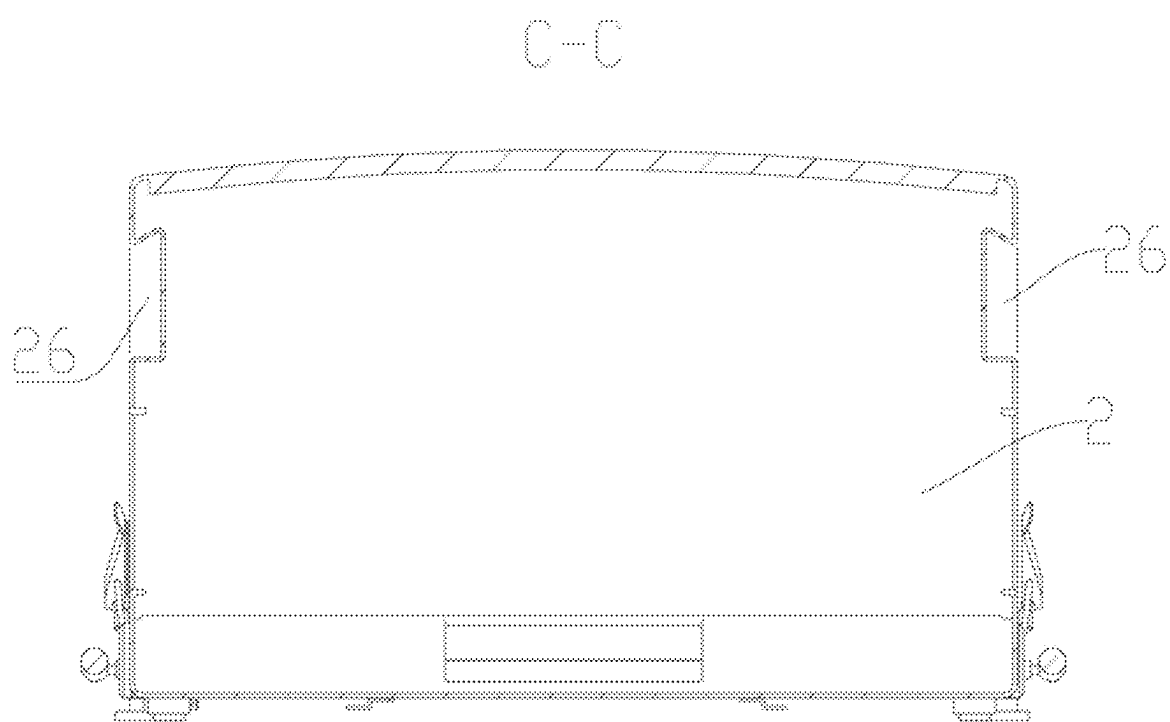


FIG. 21

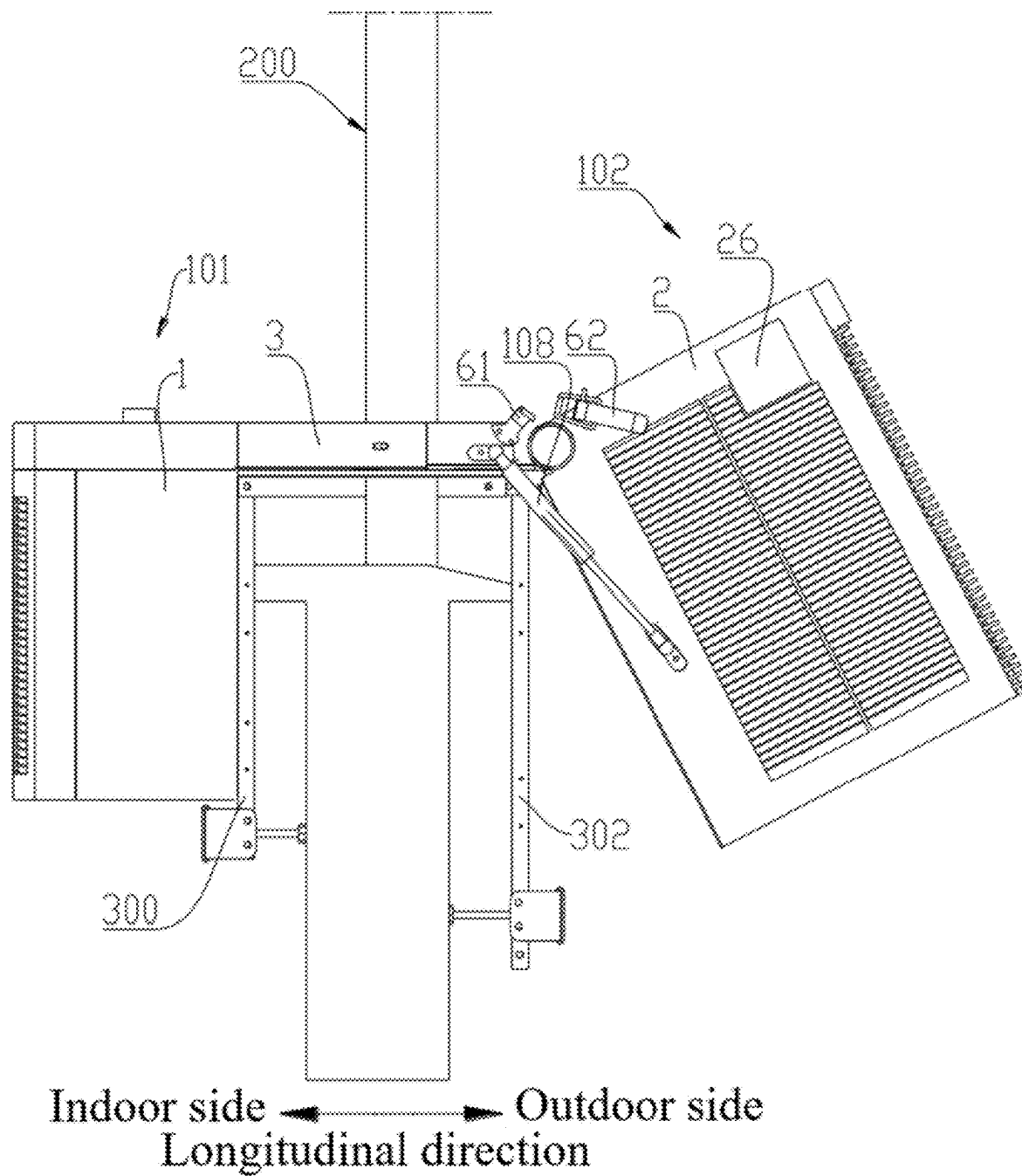


FIG. 22

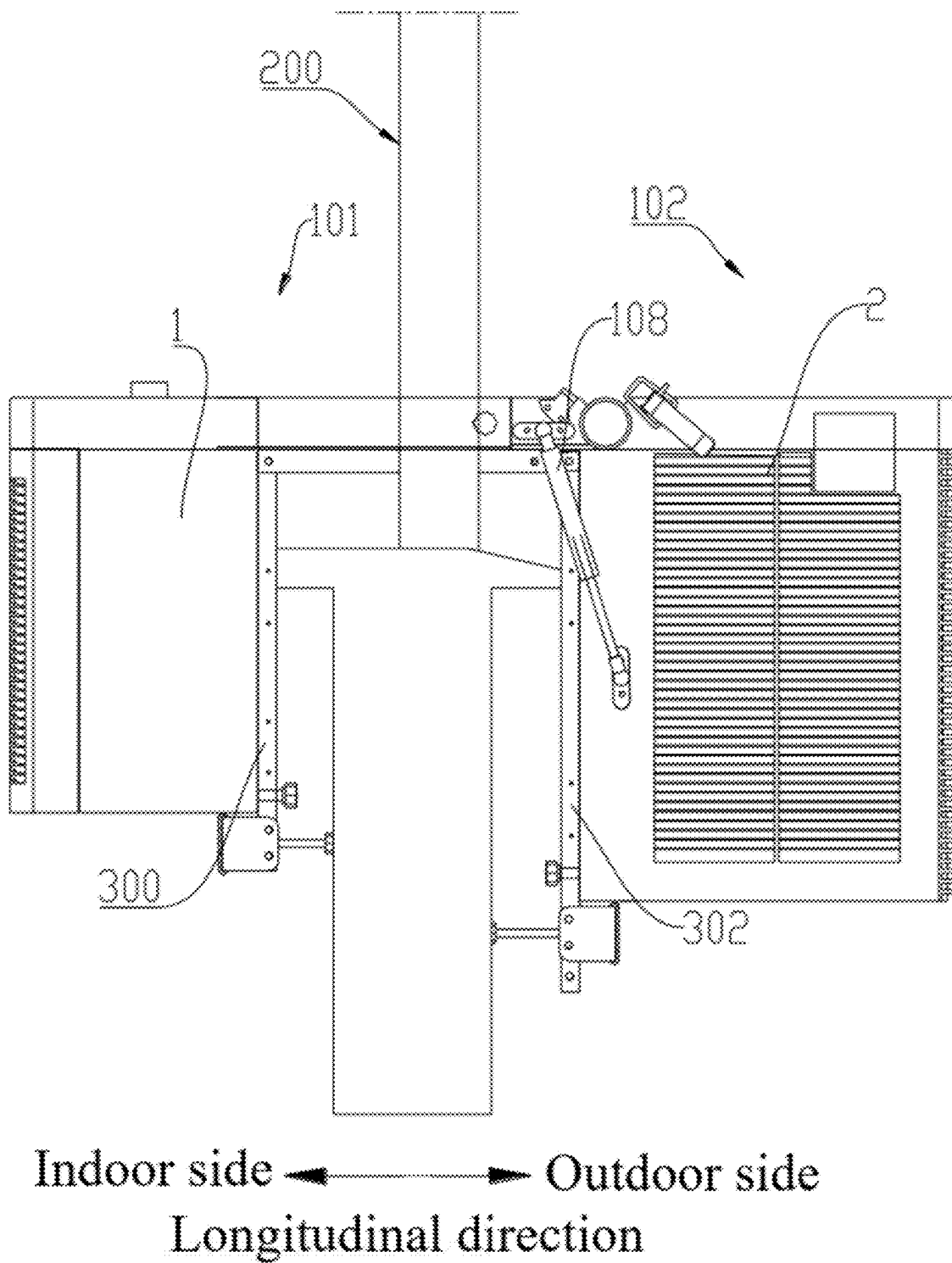


FIG. 23

WINDOW AIR CONDITIONER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is submitted based on and claims priority to Chinese Patent Application Nos. 202210602513.4 and 202221330638.8, both filed on May 30, 2022, the entire disclosures of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to the field of air conditioning technologies, and more particularly, to a window air conditioner.

BACKGROUND

[0003] A window air conditioner in the related art is an integrated air conditioner that may be mounted for use at a window opening. In order to meet noise reduction requirements, some window air conditioners are designed in a form of a saddle with an open bottom groove between an outdoor unit and an indoor unit, in order to allow the window air conditioners to be stuck on a window sill using the groove and block noise of an outdoor unit by a physical wall. However, this type of window air conditioner has a fixed configuration that is difficult to meet requirements except for the use.

SUMMARY

[0004] The present disclosure aims to solve at least one of the technical problems in the related art. To this end, the present disclosure is to provide a window air conditioner, capable of changing its configuration to meet other requirements besides use.

[0005] A window air conditioner according to embodiments of the present disclosure includes an indoor unit component and an outdoor unit component. The indoor unit component includes an indoor unit body. The outdoor unit component includes an outdoor unit body. The outdoor unit body and the indoor unit body of the window air conditioner are spaced apart from each other in a longitudinal direction, to enable the window air conditioner to have a use configuration in which the indoor unit body is located on an indoor side and the outdoor unit body is located on an outdoor side. The indoor unit component is rotatably connected to the outdoor unit component, to enable one of the outdoor unit component and the indoor unit component to rotate relative to another one of the outdoor unit component and the indoor unit component about a rotation axis extending in a transverse direction. Therefore, the window air conditioner according to the embodiments of the present disclosure can change its configuration to meet other requirements besides use.

[0006] In some embodiments, the outdoor unit body is reciprocally rotatable about the rotation axis between a first state and a second state. In the first state, a back plate of the outdoor unit body is vertically arranged, and the rotation axis is located at a position at which a top portion of the back plate of the outdoor unit body is located. In the second state, the back plate of the outdoor unit body is transversely arranged.

[0007] In some embodiments, the indoor unit component further includes a connection support adapted to pass

through a window opening. The connection support has an outer end extending to be pivotally connected to an upper inner end of the outdoor unit body, to enable the rotation axis to be located at an intersection between the back plate and a top plate of the outdoor unit body. The back plate of the outdoor unit body is flush with a bottom plate of the connection support when the outdoor unit body is in the second state.

[0008] In some embodiments, the back plate of the outdoor unit body has a first guide structure. The first guide structure is adapted to be engaged with a guide rail at a mounting support to allow a drawably guiding of the outdoor unit body in an inward-outward direction in the second state, and the mounting support is mounted at the window opening.

[0009] In some embodiments, the indoor unit component further includes a connection support adapted to pass through a window opening. The connection support has an outer end extending to be pivotally connected to an upper inner end of the outdoor unit body, to enable the rotation axis to be located at an intersection between the back plate and a top plate of the outdoor unit body. The connection support has a second guide structure at a bottom of the connection support, the second guide structure being adapted to be engaged with the guide rail after the first guide structure.

[0010] In some embodiments, the window air conditioner further includes a first limit structure. The first limit structure is adapted to prevent the outdoor unit body from further rotating in a current rotation direction when the outdoor unit body rotates into the second state relative to the indoor unit component.

[0011] In some embodiments, the window air conditioner further includes a second limit structure. The second limit structure is adapted to prevent, when the outdoor unit body rotates into the second state relative to the indoor unit component, the outdoor unit body from rotating reversely in a direction along which the outdoor unit body returns into the first state.

[0012] In some embodiments, the second limit structure includes a first buckle and a second buckle. The first buckle is arranged at the indoor unit component. The second buckle is arranged at the outdoor unit component. When the outdoor unit body is in the second state, the first buckle is engaged with the second buckle for locking to prevent the outdoor unit body from rotating reversely in the direction along which the outdoor unit body returns into the first state. The outdoor unit body is out of the second state when the first buckle is disengaged from the second buckle for unlocking.

[0013] In some embodiments, the second limit structure further includes a pull rod and a slide groove extending in the longitudinal direction and formed at the indoor unit component. The slide groove has a locking groove at an inner end of the slide groove, and the locking groove extends upwardly. The pull rod has two ends comprising a first end rotatably connected to the outdoor unit body and a second end engaged with the slide groove to be slidable along the slide groove. The second limit structure is configured to prevent the outdoor unit body from rotating reversely in the direction along which the outdoor unit body returns into the first state when the second end is engaged with the locking groove.

[0014] In some embodiments, the indoor unit component is pivotally connected to the outdoor unit component.

[0015] In some embodiments, the outdoor unit body has a hand grip groove. The hand grip groove is located on at least one of a transverse side plate, the top plate, or the back plate of the outdoor unit body.

[0016] Additional aspects and advantages of the present disclosure will be provided at least in part in the following description, or will become apparent in part from the following description, or can be learned from the practice of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a perspective view of a window air conditioner in a use configuration according to an embodiment of the present disclosure.

[0018] FIG. 2 is a view showing a use state of the window air conditioner illustrated in FIG. 1 in the use configuration.

[0019] FIG. 3 is a side view of the window air conditioner illustrated in FIG. 1 in a mounting configuration.

[0020] FIG. 4 is a view showing a mounting state of the window air conditioner illustrated in FIG. 3 in the mounting configuration.

[0021] FIG. 5 is a side view of the window air conditioner illustrated in FIG. 1 in an intermediate configuration.

[0022] FIG. 6 is a side view of the window air conditioner illustrated in FIG. 4 in a mounting configuration.

[0023] FIG. 7 is a perspective view of the window air conditioner illustrated in FIG. 6.

[0024] FIG. 8 is a view showing an engagement state between the window air conditioner illustrated in FIG. 7 and a mounting support.

[0025] FIG. 9 is another view showing an engagement state between the window air conditioner illustrated in FIG. 7 and a mounting support.

[0026] FIG. 10 is a schematic view of an internal structure of a window air conditioner according to an embodiment of the present disclosure.

[0027] FIG. 11 is a partial cross-sectional view of a window air conditioner in a rotation position according to an embodiment of the present disclosure.

[0028] FIG. 12 is a perspective view of a window air conditioner in a mounting configuration according to an embodiment of the present disclosure.

[0029] FIG. 13 is a partial enlarged view of part B illustrated in FIG. 12.

[0030] FIG. 14 is a partial enlarged view of part A illustrated in FIG. 1.

[0031] FIG. 15 is an assembly view of a part of the window air conditioner according to an embodiment of the present disclosure.

[0032] FIG. 16 is a schematic view of the window air conditioner illustrated in FIG. 15 in a mounting configuration.

[0033] FIG. 17 is a cross-sectional view of the window air conditioner illustrated in FIG. 16.

[0034] FIG. 18 is an assembly view of a part of a window air conditioner according to an embodiment of the present disclosure.

[0035] FIG. 19 is an assembly view of a part of a window air conditioner according to an embodiment of the present disclosure.

[0036] FIG. 20 is a schematic view of a part of a window air conditioner according to an embodiment of the present disclosure.

[0037] FIG. 21 is a cross-sectional view taken along line C-C illustrated in FIG. 20.

[0038] FIG. 22 is a view showing a mounting state of a window air conditioner according to an embodiment of the present disclosure.

[0039] FIG. 23 is a view showing a state in which the window air conditioner illustrated in FIG. 22 is mounted in place.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0040] The embodiments of the present disclosure will be described in detail below with reference to examples thereof as illustrated in the accompanying drawings, throughout which same or similar elements, or elements having same or similar functions, are denoted by same or similar reference numerals. The embodiments described below with reference to the drawings are illustrative only, and are intended to explain rather than limit the present disclosure. Many different embodiments or examples according to the present disclosure are used.

[0041] to realize different structures of the present disclosure. To simplify the present disclosure, components and settings in specific examples are described below. Of course, they are merely exemplary and are not intended to limit the present disclosure. Moreover, the present disclosure may repeat reference numbers and/or reference letters in different examples. Such repetition is for purposes of simplicity and clarity and is not in itself indicative of a relationship among the various embodiments and/or settings discussed. In addition, the present disclosure provides examples of various specific processes and materials, but those skilled in the art may recognize application of other processes and/or use of other materials.

[0042] A window air conditioner 100 according to an embodiment of the present disclosure is described below with reference to the drawings.

[0043] As illustrated in FIGS. 1 and 2, a window air conditioner 100 includes an indoor unit component 101 and an outdoor unit component 102. The indoor unit component 101 includes an indoor unit body 1. The outdoor unit component 102 includes an outdoor unit body 2. The outdoor unit body 2 and the indoor unit body 1 of the window air conditioner 100 are spaced apart from each other in a longitudinal direction, to enable the window air conditioner 100 to have a use configuration in which the indoor unit body 1 is located on an indoor side and the outdoor unit body 2 is located on an outdoor side. It should be noted that the window air conditioner 100 described herein is adapted to be arranged for use at a window opening 200. An inward-outward direction of the window opening 200 (i.e., a direction passing through the window opening 200) is a “longitudinal direction”; a width direction of the window opening 200 is a “transverse direction”; and a height direction of the window opening 200 is a “vertical direction.”

[0044] In short, in the window air conditioner 100 according to the embodiment of the present disclosure, the indoor unit component 101 includes the indoor unit body 1 adapted to be arranged on the indoor side, and the outdoor unit component 102 includes the outdoor unit body 2 adapted to be arranged on the outdoor side. The indoor unit body 1 and the outdoor unit body 2 are spaced apart from each other in an inward-outward direction when the window air conditioner 100 is in the use configuration (such as a configuration

illustrated in FIGS. 1 and 2). In this case, the indoor unit body 1 is arranged at the indoor side for adjusting an indoor environment temperature and the like, and the outdoor unit body 2 is arranged at the outdoor side for heat exchange with an outdoor environment.

[0045] In some optional examples, the indoor unit body 1 may include an indoor side heat exchanger, an indoor side fan, etc., and the outdoor unit body 2 may include a compressor, an outdoor side heat exchanger, an outdoor side fan, etc. The indoor unit body 1 is connected to the outdoor unit body 2 via a refrigerant pipe, and therefore the indoor side heat exchanger, the outdoor side heat exchanger, the compressor, etc. constitute a refrigerant circulation system to realize a refrigeration cycle or a heating cycle. Of course, the present disclosure is not limited to thereto. For example, in some other embodiments of the present disclosure, the indoor side fan, the outdoor side fan, etc. may also be omitted, and no examples are given herein.

[0046] As illustrated in FIG. 1, the indoor unit component 101 is rotatably connected to the outdoor unit component 102, to enable one of the outdoor unit component 102 and the indoor unit component 101 to rotate relative to another one of the outdoor unit component 102 and the indoor unit component 101 about a rotation axis L extending in a transverse direction, and thus to enable the window air conditioner 100 to change its configuration. It should be noted that the expression “transverse direction” in “the rotation axis L extending in the transverse direction” described herein refers to a transverse direction when the window air conditioner 100 is in the use configuration.

[0047] Thus, when one of the outdoor unit component 102 and the indoor unit component 101 serves as a rotating component and another of the outdoor unit component 102 and the indoor unit component 101 serves as a stationary component, and when it is necessary to change the configuration of the window air conditioner 100, force may be applied to enable the rotating component to rotate relative to the stationary component about the rotation axis L. Therefore, the configuration of the window air conditioner 100 can be changed to meet different needs. For example, mounting requirements, handling requirements, packaging requirements, transportation requirements, or the like of the window air conditioner 100 are met by changing the configuration of the window air conditioner 100. As a result, the window air conditioner 100 does not need to be restricted by the use configuration of the window air conditioner 100 (for example, the configuration illustrated in FIGS. 1 and 2) in scenarios such as mounting, handling, packaging, transportation, or the like, and can realize these scenarios by flexibly changing the configuration of the window air conditioner 100 (for example, changing into a configuration illustrated in FIG. 3 or FIG. 4).

[0048] In some embodiments, the indoor unit component 101 is pivotally connected to the outdoor unit component 102, and therefore the rotation axis L about which the outdoor unit component 102 and the indoor unit component 101 rotate relative to each other is uniquely determined instead of having a plurality of rotation axes or other situations. Thus, a rotation trajectory of the rotating component relative to the stationary component is determined. Therefore, when it is necessary to change the configuration of the window air conditioner 100, the rotating component may be allowed to rotate relative to the stationary component about the uniquely determined rotation axis L, thus

allowing the rotation trajectory of the rotating component to be unique and determined. In this way, the rotation of the rotating component is smooth, and a rotation support is reliable (i.e., supported at the unique rotation axis L), thereby effectively ensuring that the window air conditioner 100 can reliably change its configuration, and an action of driving the rotating component to rotate is simple, smooth, and labor-saving. In some optional examples, the indoor unit component 101 may be pivotally connected to the outdoor unit component 102 by a pivot mechanism such as a hinge assembly 106, a bearing, and a rotating shaft.

[0049] It should be noted that if the pivot connection about the unique rotation axis L is replaced by a movable connection of at least one connection rod, there are at least two rotation axes. When the rotating component rotates relative to the stationary component, a movement trajectory of the rotating component is undetermined, resulting in a difficulty in an operation of driving the rotating component to rotate and poor control. Moreover, if there are a plurality of rotation axes, it is difficult to reliably and effectively support the rotating component through the rotation position. For example, when the indoor unit component is connected to the outdoor unit component by a connection rod rotatable at two ends, it is difficult for the connection rod to form a reliable support on the rotating component, thus leading to a more difficulty in driving the rotating component to move.

[0050] In some embodiments, the outdoor unit body 2 is reciprocally rotatable about a rotation axis L between a first state (such as a state illustrated in FIGS. 1 and 2) and a second state (such as a state illustrated in FIGS. 3 and 4). As illustrated in FIGS. 1 and 2, in the first state, a back plate (i.e., a first back plate 21) of the outdoor unit body 2 is vertically arranged, and the rotation axis L is located at a position at which a top portion of the back plate of the outdoor unit body 2 is located (but a longitudinal position of the rotation axis L is not limited). As illustrated in FIGS. 3 and 4, in the second state, the back plate of the outdoor unit body 2 (i.e., the first back plate 21) is transversely arranged to serve as a bottom wall of the outdoor unit body 2.

[0051] It should be noted that the expression “vertically arranged” described herein refers to a vertical or substantially vertical orientation, and the expression “transversely arranged” described herein refers to a horizontal or substantially horizontal orientation, which should be understood in a broad sense. In addition, it should be noted that the back plate (i.e., the first back plate 21) of the outdoor unit body 2 refers to a structure of the outdoor unit body 2 at a side of the outdoor unit body 2 facing towards a wall at a window opening when the window air conditioner 100 is in the use configuration. For example, when the outdoor unit body 2 has a closed structure, the first back plate 21 may be a side wall surface of a housing of the outdoor unit body 2. For another example, when the outdoor unit body 2 has a semi-open structure, the first back plate 21 may also be a side wall surface of a condenser.

[0052] In addition, it should be noted that “the outdoor unit body 2 is reciprocally rotatable about the rotation axis L between the first state and the second state” is intended to illustrate that the outdoor unit body 2 has an ability to switch between the above-mentioned two states through the rotation, but it is not limited to achieve the switching of the above-mentioned two states by driving the outdoor unit body 2 to rotate necessarily. For example, when it is necessary to switch the state of the outdoor unit body 2, it can

be achieved by driving the outer unit component **102** to rotate or by driving the indoor unit component **101** to rotate, which falls within the protection scope of the present disclosure.

[0053] For example, when the window air conditioner **100** is in the use configuration (as illustrated in FIGS. **1** and **2**, for example), the outdoor unit body **2** may be changed into the first state. When the window air conditioner **100** needs to be changed to a mounting configuration for ease of mounting (as illustrated in FIGS. **3** and **4**, for example), the outdoor unit body **2** may be changed into the second state. In addition, it should be noted that when the window air conditioner **100** is switched between the use configuration and the mounting configuration, the position and the state of the indoor unit body **1** may be changed or not changed, which are not limited herein.

[0054] It can be understood that a vertical height position of the rotation axis **L** may be maintained unchanged whether the outdoor unit body **2** is in the first state or in the second state. When the outdoor unit body **2** is in the first state, the rotation axis **L** is located at a position at which a top portion of the outdoor unit body **2** is located. When the outdoor unit body **2** is in the second state, since the back plate of the outdoor unit body **2** is raised to a state of being transversely arranged, the rotation axis **L** is equivalent to being located at a position at which a bottom portion of the outdoor unit body **2** is located.

[0055] That is, it is roughly equivalent to: when the outdoor unit body **2** is in the first state, the outdoor unit body **2** as a whole is generally located at lower level than a horizontal plane where the rotation axis **L** is located; and when the outdoor unit body **2** is in the second state, the outdoor unit body **2** as a whole is generally located at higher level than the horizontal plane where the rotation axis **L** is located. For example, the outdoor unit body **2** may rotate upwardly with a bottom raised (in a counterclockwise direction illustrated in FIG. **3**), enabling the outdoor unit body **2** to change from the first state (the state illustrated in FIGS. **1** and **2**) into the second state (the state illustrated in FIGS. **3** and **4**).

[0056] Thus, when the outdoor unit body **2** changes from the first state into the second state, since the outdoor unit body **2** is raised relative to the rotation axis **L** as a whole, the outdoor unit body **2** can be easily pushed out of the window opening **200** from the indoor side to the outdoor side. Therefore, a difficulty of the mounting of the window air conditioner **100** is reduced. As a result, the mounting of the window air conditioner **100** is more labor-saving. That is, it is avoided that the operation of pushing the outdoor unit body **2** from the inside out by raising the window air conditioner **100** as a whole to enable the bottom wall of the outdoor unit body **2** to be located at higher level than a windowsill. As a result, the operation is more labor-saving. Moreover, since there is no need to raise the window air conditioner **100** as a whole and then push out the outdoor unit body **2**, a risk of the whole machine tipping over and falling off towards the outdoor side, which is caused by relatively high height of a center of gravity of the whole machine and the difficulty in controlling the whole machine when pushing the whole machine from the inside out, is avoided. Thus, safety of the mounting is improved.

[0057] For example, as illustrated in FIGS. **1** and **2**, when the window air conditioner **100** is in the use configuration, the indoor unit body **1** and the outdoor unit body **2** are

spaced apart from each other in an inward-outward direction. In this case, a bottom plate (i.e., a second bottom plate **12**) of the indoor unit body **1** faces downwards; a top plate (i.e., a second top plate **13**) of the indoor unit body **1** faces upwards; a panel (i.e., a second panel **14**) of the indoor unit body **1** faces towards the indoor side; and a back plate (i.e., a second back plate **11**) of the indoor unit body **1** faces towards the outdoor side. A bottom plate (i.e., a first bottom plate **22**) of the outdoor unit body **2** faces downwards; a top plate (i.e., a first top plate **23**) of the outdoor unit body **2** faces upwards; a panel (i.e., the first panel **24**) of the outdoor unit body **2** faces towards the outdoor side; and a back plate (i.e., a first back plate **21**) of the outdoor unit body **2** faces towards the indoor side. The outer unit component **102** has an upper inner end pivotally connected to an upper outer end of the indoor unit component **101**.

[0058] For example, as illustrated in FIGS. **3** and **4**, if the outdoor unit component **102** is pulled upwards to enable the outdoor unit component **102** to counterclockwise pivot about the rotation axis **L**, the window air conditioner **100** is in the mounting configuration after the outdoor unit component **102** rotates by 90°. In this case, the bottom plate (i.e., the first bottom plate **22**) of the outdoor unit body **2** faces the outdoor side, the top plate (i.e., the first top plate **23**) of the outdoor unit body **2** faces towards the indoor side, the panel (i.e., the first panel **24**) of the outdoor unit body **2** faces upwards, and the back plate (i.e., the first back plate **21**) of the outdoor unit body **2** faces downwards. The indoor unit body **1** still maintains the bottom plate (i.e., the second bottom plate **12**) facing downwards, the top plate (i.e., the second top plate **13**) facing upwards, the panel (i.e., the second panel **14**) facing towards the indoor side, and the back plate (i.e., the second back plate **11**) facing towards the outdoor side.

[0059] For example, as illustrated in FIG. **5**, if the indoor unit component **101** is pulled upwards to enable the indoor unit component **101** to clockwise pivot about the rotation axis **L**, the window air conditioner **100** is in an intermediate configuration after the indoor unit component **101** rotates by 90°. In this case, the bottom plate (i.e., the second bottom plate **12**) of the indoor unit body **1** faces towards the indoor side; the top plate (i.e., the second top plate **13**) of the indoor unit body **1** faces towards the outdoor side; the panel (i.e., the second panel **14**) of the indoor unit body **1** faces upwards; and the back plate (i.e., the second back plate **11**) of the indoor unit body **1** faces downwards. The outdoor unit body **2** still maintains the bottom plate (i.e., the first bottom plate **22**) facing downwards, the top plate (i.e., the first top plate **23**) facing upwards, the panel (i.e., the first panel **24**) facing towards the outdoor side, and the back plate (i.e., the first back plate **21**) facing towards the indoor side. It can be understood that the window air conditioner **100** may be in the mounting configuration (as illustrated in FIG. **3**, for example) by counterclockwise rotating the window air conditioner **100** in the intermediate configuration (as illustrated in FIG. **5**) by 90° as a whole.

[0060] In summary, as illustrated in FIGS. **1** and **2**, when the window air conditioner **100** is in the use configuration, the rotation axis **L** is located at the position at which the top portion of the outdoor unit body **2** is located. As illustrated in FIGS. **3** and **4**, when the window air conditioner **100** is in the mounting configuration, the rotation axis **L** is located at the position at which the bottom portion of the outdoor unit body **2** is located. Since the vertical height of the rotation

axis L remains unchanged, it is equivalent to raising the outdoor unit body 2 as a whole. Therefore, the outdoor unit body 2 can be easily pushed outwards from the indoor side to the outdoor side through the window opening 200 without changing the state of the indoor unit body 1. In this way, the difficulty of the mounting of the window air conditioner 100 is reduced. As a result, the mounting of the window air conditioner 100 is more labor-saving and easier to control, thereby reducing the risk of the whole machine tipping over and falling off towards the outdoor side.

[0061] It can be understood that if the window air conditioner 100 always maintains the use configuration, then when the outdoor unit body 2 needs to be pushed outwards from the window opening 200, the window air conditioner 100 needs to be raised as a whole, which is laborious to operate. Moreover, if the window air conditioner 100 always maintains the use configuration, when the whole machine is raised to be pushed outwards, the center of gravity of the whole machine is relatively high because the indoor unit component 101 is also located at a relatively high level (for example, higher than a bottom edge of the window opening 200). In this case, there is a problem of the outdoor unit body 2 tipping over outwards, which is difficult to control and dangerous.

[0062] In the window air conditioner 100 according to some embodiments of the present disclosure, since the indoor unit component 101, in the mounting configuration, may still maintain to be located at the same level as in the use configuration, for example, at lower level than the bottom edge of the window opening 200, an installer can easily press against the indoor unit body 1 from a top of the indoor unit body 1 to avoid the problem of the outdoor unit body 2 tipping over and falling off outwards, which is easy to control and reduces a risk.

[0063] In addition, as mentioned above, there are many ways to change the window air conditioner 100 from the use configuration to the mounting configuration. For example, as illustrated in FIGS. 1 and 3, the outdoor unit body 2 may rotate from the first state into the second state by counterclockwise rotating about the rotation axis L by 90°, to allow the window air conditioner 100 to change from the use configuration to the mounting configuration. Alternatively, for example, as illustrated in FIGS. 1 and 5, the indoor unit body 1 may clockwise rotate about the rotation axis L by 90° to allow the window air conditioner 100 to change into the intermediate configuration. With reference to FIGS. 3 and 5, the whole machine counterclockwise rotates by 90° to allow the window air conditioner 100 to change from the intermediate configuration to the mounting configuration (such as the configuration illustrated in FIG. 3).

[0064] In some embodiments, as illustrated in FIGS. 1 and 2, the indoor unit component 101 includes a connection support 3 adapted to pass through the window opening 200. The connection support 3 has an outer end extending to be pivotally connected to an upper inner end of the outdoor unit body 2, to enable the rotation axis L to be located at an intersection between the back plate (i.e., the first back plate 21) and the top plate (i.e., the first top plate 23) of the outdoor unit body 2 (i.e., a position of a top portion of the outdoor unit body 2 close to the indoor side when the window air conditioner 100 is in the use configuration).

[0065] It should be noted that “the outer end of the connection support 3 extends to be pivotally connected to the upper inner end of the outdoor unit body 2” is intended

to illustrate the connection position between the indoor unit component 101 and the outdoor unit body 2, and does not limit how to achieve the connection, for example, which may be a direct connection or an indirect connection. Moreover, a setting position of a linkage for the indirect connection is not limited, for example, which may be arranged at the connection support 3, or may be arranged at the indoor unit body 1.

[0066] Therefore, since the indoor unit component 101 includes a connection support 3 that may pass through the window opening 200 and is pivotally connected to the outdoor unit body 2, the rotation axis L may be located at the upper inner end of the outdoor unit body 2 (i.e., the intersection between the back plate and the top plate of the outdoor unit body 2). In this way, when the indoor unit component 101 serves as a stationary component and the outdoor unit component 102 is rotated, it is equivalent to that only the outdoor unit body 2 rotates about the rotation axis L at the upper inner end of the outdoor unit body 2. Therefore, this arrangement can provide a reliable rotation support at the rotation axis L to improve stability and reliability of the pivoting of the outdoor unit body 2. Moreover, this arrangement can reduce a space swept by the rotation of the outdoor unit component 102 as a whole as well as drive torque required to drive the outdoor unit component 102 to rotate. As a result, the operation is more labor-saving, and a height of the window opening 200 is required to be lower (for example, the height of the window opening 200 may be lower).

[0067] It should be noted that the connection relation between the indoor unit body 1 and the connection support 3 is not limited. For example, the indoor unit body 1 and the connection support 3 may be fixedly connected to each other or may be slidably connected to each other in such a manner that the indoor unit body 1 and the connection support 3 move relative to each other in the longitudinal direction, which is not limited herein. When the indoor unit body 1 and the connection support 3 are fixedly connected to each other, at least part of the connection support 3 is always located outside the indoor unit body 1, to enable the outdoor unit body 2 to be spaced apart from the indoor unit body 1 in the longitudinal direction. When the indoor unit body 1 and the connection support 3 are relatively slidably connected to each other in the longitudinal direction, and the window air conditioner 100 is in the use configuration, at least part of the connection support 3 is located outside the indoor unit body 1, to enable the outdoor unit body 2 to be spaced apart from the indoor unit body 1 in the longitudinal direction. When the indoor unit body 1 and the connection support 3 are relatively slidably connected to each other in the longitudinal direction, and the window air conditioner 100 is in the mounting configuration, the connection support 3 may be stacked on a top portion of the indoor unit body 1, to enable the indoor unit component 1 to be adjacent to the outer unit component 2, or at least part of the connection support 3 is located outside the indoor unit body 1, to enable the indoor unit body 1 to be spaced apart from the outdoor unit body 2 in the longitudinal direction.

[0068] When the indoor unit body 1 and the connection support 3 are slidably connected to each other in such a manner that the indoor unit body 1 and the connection support 3 move relative to each other in the longitudinal direction, a relative longitudinal position of the outdoor unit body 2 and the indoor unit body 1 may be adjusted, which

not only helps to reduce a longitudinal distance between the outdoor unit body 2 and the indoor unit body 1 for ease of packaging and transportation, but also allows the longitudinal distance between the outdoor unit body 2 and the indoor unit body 1 to match longitudinal dimension requirements of different windowsills.

[0069] Of course, the present disclosure is not limited thereto. For example, in other embodiments of the present disclosure, the connection support 3 may be replaced with a first support and a second support. The indoor unit component 101 includes a first support extendable into the window opening 200 from the inside out. The outdoor unit component 102 includes a second support extendable into the window opening 200 from the outside in. The first support has an outer end (i.e., an end close to the outdoor side) pivotally connected to an inner end of the second support (i.e., an end close to the indoor side) (this embodiment is not shown in the figures). At this time, when the indoor unit component 101 serves as a stationary component and the outdoor unit component 102 is rotated, the outdoor unit body 2 rotates together with the second support about the rotation axis L, and a space swept by the rotation of the outdoor unit component 102 as a whole is relatively large. As such, drive torque required to drive the outdoor unit component 102 to rotate is large, and a height of the window opening 200 is required to be slightly higher (for example, the height of the window opening 200 needs to be slightly higher).

[0070] Alternatively, in some other embodiments of the present disclosure, the connection support 3 may be replaced with a third support. At this time, the outdoor unit component 102 includes a third support adapted to pass through the window opening 200. The third support has an inner end (i.e., an end close to the indoor side) pivotally connected to an upper outer end of the indoor unit body 1 (i.e., an end of the top portion of the indoor unit body 1 close to the outdoor side when the window air conditioner 100 is in the use configuration) (this embodiment is not shown in the figures). At this time, when the indoor unit component 101 serves as a stationary component and the outdoor unit component 102 is rotated, the outdoor unit body 2 rotates together with the third support about the rotation axis L, and a space swept by the rotation of the outdoor unit component 102 as a whole is a larger. As such, drive torque required to drive the outdoor unit component 102 to rotate is larger, and a height of the window opening 200 is required to be higher (for example, the height of the window opening 200 needs to be higher, especially a higher height for the window opening 200 is required as the longitudinal dimension of the window opening 200 becomes larger).

[0071] In addition, when the indoor unit component 101 includes a connection support 3, and the connection support 3 has an outer end pivotally connected to an upper inner end of the outdoor unit body 2, compared with the solution in which the outdoor unit component 102 further includes a support section (such as the second support or the third support mentioned above), when the outdoor unit component 102 is rotated, the outdoor unit body 2 is prevented from applying a force to the support section (such as the second support or the third support mentioned above), which triggers hidden trouble such as damage and fracture of the support section under pressure. Thus, structural reliability is improved at a low cost. Furthermore, when the outdoor unit component 102 includes no support section, the structure of

the outdoor unit component 102 can be simplified, the cost can be reduced, and the assembly can be simplified.

[0072] In some embodiments, as illustrated in FIG. 6, when the outdoor unit body 2 is in the second state, the back plate (i.e., the first back plate 21) of the outdoor unit body 2 is flush with a bottom plate of the connection support 3. It should be noted that the expression “flush” herein may refer to completely flush or roughly flush. Thus, when the outdoor unit body 2 is pushed from the indoor side to the outdoor side, the window air conditioner 100 as a whole hardly moves in the vertical direction in a process of the outdoor unit body 2 passing through the window opening 200, and the connection support 3 can immediately follow the outdoor unit body 2 and also pass through the window opening 200, thereby simplifying the operation. As a result, the operation is more labor-saving and convenient, and assembly efficiency is higher.

[0073] In some embodiments, as illustrated in FIGS. 6 to 8, the back plate (i.e., the first back plate 21) of the outdoor unit body 2 has a first guide structure 25. The first guide structure 25 is adapted to be engaged with a guide rail 301 at a mounting support 300 to allow a drawable guiding of the outdoor unit body 2 in an inward-outward direction in the second state, the mounting support 300 is mounted at the window opening 200. Thus, when the window air conditioner 100 is in the mounting configuration, the outdoor unit body 2 may be smoothly and labor-savingsly pushed outwards from the indoor side to the outdoor side using the engagement between the first guide structure 25 and the guide rail 301.

[0074] Further, with reference to FIGS. 6, 7, and 9, the connection support 3 has a second guide structure 31 at a bottom of the connection support 3. The second guide structure 31 is adapted to be engaged with the guide rail 301 after the first guide structure 25. That is, when the window air conditioner 100 is in the mounting configuration, first, the outdoor unit body 2 may be smoothly and labor-savingsly pushed outwards from the indoor side to the outdoor side using the engagement between the first guide structure 25 and the guide rail 301, and then the connection support 3 may be pushed into the window opening 200 using the engagement between the second guide structure 31 and the guide rail 301. In this process, the window air conditioner 100 as a whole does not need to move in the vertical direction, and the connection support 3 can immediately follow the outdoor unit body 2 and also pass through the window opening 200, thereby simplifying the operation. As a result, the operation is more labor-saving and convenient, and the assembly efficiency is higher. It should be noted that a shape of each of the first guide structure 25, the second guide structure 31, and the guide rail 301 is not limited, as long as the shape of each of the first guide structure 25, the second guide structure 31, and the guide rail 301 may better meet guide requirements. For example, each of the first guide structure 25, the second guide structure 31, and the guide rail 301 may be an elongated lath, which is not limited herein.

[0075] In short, the second guide structure 31 and the first guide structure 25 are arranged at the bottom of the connection support 3 and the back plate of the outdoor unit body 2, respectively. Meanwhile, the guide rail 301 is arranged at the mounting support 300. The window air conditioner 100 is changed into the mounting configuration, and the indoor unit body 1 and the outdoor unit body 2 are spaced apart

from each other in the longitudinal direction and connected to each other by the connection support 3. At this time, the window air conditioner 100 is placed on the mounting support 300, to allow the first guide structure 25 at the bottom of the outdoor unit body 2 to be engaged with the guide rail 301 and the window air conditioner 100 to be pushed outwards under the guide of the guide rail 301. As the whole machine moves outwards, the first guide structure 25 is decreasingly engaged with the guide rail 301; the second guide structure 31 gradually moves to be engaged with the guide rail 301, and the second guide structure 31 is increasingly engaged with the guide rail 301. Thus, reliable cooperation between the whole machine and the mounting support 300 is ensured. When the outdoor unit body 1 fully extends to an outside of the window opening 200, the first guide structure 25 is disengaged from the guide rail 301, and the second guide structure 31 is still engaged with the guide rail 301. As a result, the whole machine can reliably cooperate with the mounting support 300. Afterwards, the outdoor unit body 1 may be turned downwards to allow the outdoor unit body 1 to be changed into the first state that meets the use requirements.

[0076] In some embodiments of the present disclosure, the back plate (i.e., the first back plate 21) of the outdoor unit body 2 may further have two reinforced guide rails arranged on two transverse sides of the back plate of the outdoor unit body 2. The first guide structures 25 each are located between the two reinforced guide rails. When the outdoor unit body 2 is in the second state, the two reinforcing guide rails are clamped on two transverse sides of the mounting support 300 or engaged with a guide structure arranged at a corresponding position at the mounting support 300, thereby further providing a guide.

[0077] In some embodiments of the present disclosure, the window air conditioner 100 may include a first limit structure 104 (with reference to FIGS. 10 and 11). The first limit structure 104 is adapted to prevent the outdoor unit body 2 from further rotating in a current rotation direction (for example, to prevent the outdoor unit body 2 illustrated in FIG. 3 from rotating in a counterclockwise direction) when the outdoor unit body 2 rotates into the second state (for example, the state illustrated in FIG. 3) relative to the indoor unit component 101. Thus, the installer can more easily change the outdoor unit body 2 into the second state to simplify the operation and improve the assembly efficiency.

[0078] It should be noted that the structure and the setting position of the first limit structure 104 are not limited, as long as the above-mentioned limit may be achieved. For example, as illustrated in FIGS. 10 and 11, when the indoor unit component 101 is pivotally connected to the outer unit component 102 by the hinge assembly 106, the first limit structure 104 may be integrated into the hinge assembly 106. For example, the hinge assembly 106 may include a first articulation member 41 and a second articulation member 42 that rotate relative to each other. The first articulation member 41 is provided with a first stop portion 411, and the second articulation member 42 is provided with a second stop portion 421. When the outer unit body 2 rotates from the first state into the second state, the first stop portion 411 cooperates with the second stop portion 421 to prevent the outer unit body 2 from further rotating in a current rotation direction.

[0079] For another example, with reference to FIG. 11, the first limit structure 104 may further include a first substructure

1041 arranged at the indoor unit component 101 and a second substructure 1042 arranged at the outdoor unit component 102. When the outdoor unit body 2 rotates into the second state relative to the indoor unit component 101, the first substructure 1041 cooperates with the second substructure 1042, to prevent the outdoor unit body 2 from further rotating in the current rotation direction.

[0080] In some embodiments of the present disclosure, as illustrated in FIG. 12, the window air conditioner 100 may further include a second limit structure 105. The second limit structure 105 is adapted to prevent, when the outdoor unit body 2 rotates into the second state relative to the indoor unit component 101, the outdoor unit body 2 from rotating reversely in a direction along which the outdoor unit body 2 returns into the first state (for example, to prevent the outdoor unit body 2 illustrated in FIG. 12 from rotating in a clockwise direction). Thus, the outdoor unit body 2 can be kept relatively stable in the second state to facilitate the mounting.

[0081] It should be noted that the structure and the setting position of the second limit structure 105 are not limited, as long as the above-mentioned limit may be achieved. For example, in some embodiments, as illustrated in FIGS. 13 and 14, the second limit structure 105 is in a form of a buckle and includes a first buckle 61 and a second buckle 62. The first buckle 61 is arranged at the indoor unit component 101, and the second buckle 62 is arranged at the outdoor unit component 102. When the outdoor unit body 2 is in the second state (for example, a state illustrated in FIGS. 12 and 13), the first buckle 61 is engaged with the second buckle 62 for locking, and the second limit structure 105 is configured to prevent the outdoor unit body 2 from rotating reversely in the direction along which the outdoor unit body 2 returns into the first state. When the first buckle 61 is disengaged from the second buckle 62 for unlocking, the outdoor unit body 2 may be out of the second state and rotates reversely in the direction along which the outdoor unit body 2 returns into the first state to change back into the first state (for example, to change to the state illustrated in FIGS. 1 and 14). Therefore, the second limit structure 105 is simple and facilitates realizing the position locking.

[0082] In another exemplary embodiment of the present disclosure, as illustrated in FIGS. 12 to 14, the window air conditioner 100 may include two second limit structure 105 in a form of a buckle, and the two second limit structure 105 may be arranged on two transverse sides of the outdoor unit component 102. When the outdoor unit component 102 is in the first state (with reference to FIGS. 1 and 14), the first buckle 61 is connected to a top portion of the indoor unit component 101 close to the back plate (i.e., the second back plate 11) of the indoor unit body 1, and extends obliquely upwards from the indoor side to the outdoor side; the second buckle 62 is connected to a top portion of the outdoor unit component 102 close to the back plate (i.e., the first back plate 21) of the outdoor unit body 2, and extends obliquely upwards from the outdoor side to the indoor side. When the outdoor unit component 102 rotates into the second state (with reference to FIGS. 12 and 13), the second buckle 62 rotates to be collinear with and hooked to the first buckle 61, to allow the second buckle 62, the first buckle 61, the outdoor unit component 102, and the indoor unit component 101 to form a triangular support limit. In this way, a relatively stable limit is formed.

[0083] In some embodiments, as illustrated in FIG. 15, the second limit structure 105 may further include a pull rod 71 and a slide groove 72. The slide groove 72 extends in the longitudinal direction and formed at the indoor unit component 101. The slide groove 72 has a locking groove 73 at an inner end of the slide groove 72, and the locking groove 73 extends upwardly. The pull rod 71 has two ends including a first end 711 rotatably connected to the outdoor unit body 2 and a second end 712 engaged with the slide groove 72 to be slidable along the slide groove 72. With reference to FIG. 16, the second limit structure 105 is configured to prevent the outdoor unit body 2 from rotating reversely in the direction along which the outdoor unit body 2 returns into the first state when the second end 712 is engaged with the locking groove 73 (for example, to prevent the outdoor unit body 2 illustrated in FIG. 16 from rotating in a clockwise direction). Therefore, the second limit structure 105 is simple and facilitates realizing the position locking.

[0084] In another exemplary embodiment disclosure of the present disclosure, as illustrated in FIGS. 15 and 16, two pull rods 71 are provided and arranged on the two transverse sides of the outdoor unit component 102. As illustrated in FIG. 15, when the outdoor unit component 102 is in the first state, the first end 711 of the pull rod 71 is connected to a top portion of the outdoor unit component 102 close to a panel (i.e., a first panel 24). As illustrated in FIG. 16, when the outdoor unit component 102 rotates towards the second state, the second end 712 of the pull rod 71 is slidable towards the indoor side and engaged into the locking groove 73. At this time, the pull rod 71, the outdoor unit component 102, and the indoor unit component 101 may be connected to form a triangle, forming a relatively stable limit. Even if the outdoor unit component 102 tends to rotate downwards under gravity, since the second end 712 of the pull rod 71 is embedded into the locking groove 73, the outdoor unit component 102 cannot slide towards the outdoor side along the slide groove 72, thereby preventing the outdoor unit component 102 from rotating towards the first state illustrated in FIG. 15.

[0085] Further, with reference to FIG. 17, the second limit structure 105 may also include a return spring 74 connected between the indoor unit component 101 and the second end 712 to apply an elastic force to the second end 712 to move towards the locking groove 73. Thus, when the outdoor unit component 102 rotates from the second state into the first state, the return spring 74 may be extended to achieve a buffering effect to some extent. Moreover, when the second end 712 of the pull rod 71 is engaged into the locking groove 73, a force may be applied to the pull rod 71 under the return spring 74 to form a more stable limit and prevent the outdoor unit component 102 from rotating reversely.

[0086] In some embodiments of the present disclosure, the outer end of the connection support 3 can be pivotally connected to the outdoor unit body 2 through a hinge assembly 106. No limitation is imposed on the type of the hinge assembly 106. In some embodiments, as illustrated in FIGS. 18 and 19, the hinge assembly 106 may be an exposed hinge a, and at least part of the exposed hinge a may be exposed outside a top wall or a bottom wall of the connection support 3. Thus, the mounting and inspection of the exposed hinge a can be simplified. For example, as illustrated in FIGS. 10 and 11, the hinge assembly 106 may also be a built-in hinge b that is not exposed outside the top wall or the bottom wall of the connection support 3. Of course,

the present disclosure is not limited to this. In other embodiments of the present disclosure, the indoor unit component 101 may also be pivotally connected to the outdoor unit component 102 in other manners, such as by a bearing or a rotating shaft, which will not be repeated herein.

[0087] In some embodiments of the present disclosure, as illustrated in FIGS. 10 and 11, the hinge assembly 106 may include a first articulation member 41 and a second articulation member 42 hinged to the first articulation member 41. The first articulation member 41 is arranged at the indoor unit component 1, and a second articulation member 42 is arranged at the outdoor unit component 2. In some embodiments, the first articulation member 41 may be pivotally connected to the second articulation member 42 by a damping rotary shaft 43. For example, a spring spacer may be applied to the damping rotary shaft 43, and rotation torque may be adjusted as desired, thereby solving a problem of the outdoor unit body 2 falling off rapidly during the rotation. Moreover, the structure of the damping rotary shaft 43 is strong enough to effectively support the outdoor unit body 2 to rotate.

[0088] In some embodiments of the present disclosure, as illustrated in FIGS. 19 to 21, the outdoor unit body 2 has a hand grip groove 26. The hand grip groove is located on at least one of a transverse side plate (i.e., the first side plate 20), the top plate (i.e., the first top plate 23), or the back plate (i.e., the first back plate 21) of the outdoor unit body 2. That is, a groove-shaped structure may be processed at the outdoor unit body 2 to serve as the hand grip groove 26, thereby facilitating the rotation of the outdoor unit body 2. It is worth noting that in order to reduce a noise impact of the window air conditioner 100 on an indoor environment, a compressor may be arranged in the outdoor unit body 2. Therefore, the outdoor unit body 2 has a relatively heavy weight. By providing the hand grip groove 26, the installer can stably and reliably operate the outdoor unit body 2 to rotate, avoiding a problem of hand slipping.

[0089] For example, as illustrated in FIGS. 20 and 21, the outdoor unit body 2 has two hand grip grooves 26, and the two hand grip grooves 26 are located at an upper part of two transverse side plates of the outdoor unit body 2. For example, as illustrated in FIG. 19, the outdoor unit body 2 has two hand grip grooves 26, and the two hand grip grooves 26 are located on the back plate (i.e., the first back plate 21) and are spaced apart from each other in the transverse direction. Therefore, both arrangements are more helpful to operate the outdoor unit body 2 to rotate by grasping the hand grip groove 26, to facilitate manual change of the configuration of the window air conditioner 100.

[0090] In another embodiment of the present disclosure, the hand grip groove 26 may have a width more than 800 mm, to meet ergonomics and ensure that it is greater than a width of a palm. As a result, the machine can be effectively lifted.

[0091] In an embodiment of the present disclosure, as illustrated in FIG. 20, the window air conditioner 100 may further include a support assembly 108. The support assembly 108 is movably connected between the connection support 3 and the outdoor unit body 2 and configured to hinder (not prevent) a flipping movement of the outdoor unit body 2 with a lowered bottom. For example, when the outdoor unit body 2 rotates downwards (for example, in a clockwise direction illustrated in FIG. 20) relative to the indoor unit component 101 to lower the bottom of the

outdoor unit body 2, resistance to rotation in this direction is provided on the outdoor unit body 2 to buffer the outdoor unit body 2 to rotate downwardly and avoid the problems such as collision or falling off of the outdoor unit body 2 caused by the rapid downward rotation of the outdoor unit body 2. In this way, reliability and safety of the mounting are improved. For another example, when the outdoor unit body 2 flips upwards, the outdoor unit body 2 is hindered from rotating downwards to a certain extent through the support assembly 108. As a result, the operation of the installer can be more labor-saving and the difficulty of operation is reduced.

[0092] For example, as illustrated in FIG. 20, the support assembly 108 may include a first support member 81 and a second support member 82 that are slidable relative to each other. An energy storage medium is provided between the first support member 81 and the second support member 82. The first support member 81 is rotatably connected to the connection support 3 at an end of the first support member 81 away from the second support member 82 and in a length direction of the first support member 81. The second support member 82 is rotatably connected to the outdoor unit body 2 at an end of the second support member 82 away from the first support member 81 and in a length direction of the second support member 82. Thus, the support assembly 108 may be extended and retracted by relative sliding of the first support member 81 and the second support member 82, and supported by the energy storage medium. For example, when the outdoor unit body 2 rotates downwards, the first support member 81 and the second support member 82 move relative to each other to shorten the support assembly 108. In this case, the energy storage medium is compressed. The energy storage medium can provide the above resistance to buffer the outdoor unit body 2 to rotate downwards.

[0093] In another exemplary embodiment of the present disclosure, one of the first support member 81 and the second support member 82 is a sleeve; another one of the first support member 81 and the second support member 82 is an insert rod, and the insert rod can be pushed and pulled relative to the sleeve; and the energy storage medium is gas, or liquid, or a spring 83 provided in the sleeve. That is, the first support assembly 8 may be a pneumatic rod, a hydraulic rod, a retractable spring rod, etc. Thus, through the structural design described above, stability and reliability of the relative sliding of the first support member 81 and the second support member 82 can be guaranteed. Moreover, there are many optional types of energy storage media, which can more reliably and stably provide the support and the hindering.

[0094] In the window machine of the saddle-type in the related art, the indoor unit and the outdoor unit are relatively fixed, and the upper end of the indoor unit is connected to the upper end of the outdoor unit. During the mounting, it is necessary to raise the window machine of the saddle-type as a whole and push the outdoor unit outside the window. This operation is more laborious, and there is the risk of the outdoor unit falling off outwards.

[0095] In the window air conditioner 100 according to the embodiments of the present disclosure, the outdoor unit body 2 is pivotally arranged, which can effectively solve the technical problems described above. For example, in some embodiments of the present disclosure, the outdoor unit body 2 may rotate about the only rotation axis L by 90° from the first state (such as the state illustrated in FIGS. 1 and 2)

into the second state (such as the state illustrated in FIGS. 3 and 4). For example, the indoor unit component 101 can be lifted by hand and rotated by 90°, and the indoor unit component 101 may rotate to a vertical state (for example, as illustrated in FIG. 5); or the outdoor unit body 2 may be lifted by hand and rotated by 90°, and the outdoor unit body 2 can be rotated to a transverse position (for example, as illustrated in FIG. 3).

[0096] As illustrated in FIGS. 3 and 4, when the outdoor unit body 2 is changed into the second state, the outdoor unit body 2 is locked through the engagement between the first buckle 61 and the second buckle 62. When the indoor unit component 101 or the outdoor unit body 2 is rotated, the outdoor unit body 2 is easily rotated to by 90° by means of the support resistance of the support assembly 108.

[0097] As illustrated in FIGS. 3 and 4, when the outdoor unit body 2 rotates 90° to reach the second state, a lower bottom surface (i.e., the first back plate 21) of the outdoor unit body 2 is flush with a lower bottom surface of the connection support 3. As a result, the outdoor unit body 2 and the connection support 3 can be smoothly pushed outwards.

[0098] When the outdoor unit body 2 reaches the outdoor side, with reference to FIG. 22, the first buckle 61 is disengaged from the second buckle 62 for unlocking, and then the outdoor unit body 2 can slowly rotate and drop down under the support assembly 108. With reference to FIG. 23, when the outdoor unit body 2 rotates by 90° back into the first state, the outdoor unit body 2 may fall on a support leg 302 of the mounting support 300, and the window air conditioner 100 is supported by the mounting support 300 to allow for the safe and reliable mounting and fixing of the window air conditioner 100.

[0099] Thus, the difficulty of the mounting of the window machine of the saddle-type and the problem that the outdoor unit is too heavy to be directly lifted from the window opening 200 to the outdoor side for the mounting can be effectively solved. In addition, the window air conditioner 100 according to some embodiments of the present disclosure can realize a low-temperature heating function and can operate normally at -30° C., and details thereof will not be described herein.

[0100] In the description of the present disclosure, it is to be understood that, the terms such as “longitudinal,” “transverse,” and “length” refer to the directions and location relations which are the directions and location relations shown in the drawings, and for describing the present disclosure and for describing in simple, and which are not intended to indicate or imply that the device or the elements are arranged to locate at the specific directions or are structured and performed in the specific directions, which could not to be understood to the limitation of the present disclosure.

[0101] In addition, the terms such as “first” and “second” are used herein for purposes of description and are not intended to indicate or imply relative importance, or to implicitly show the number of technical features indicated. Thus, a feature associated with “first” and “second” may comprise one or more this feature distinctly or implicitly. In the description of the present disclosure, the “plurality of” means two or more than two, unless specified otherwise.

[0102] In the present disclosure, unless specified or limited otherwise, the terms “mounted,” “connected,” “coupled” and “fixed” are understood broadly, such as fixed,

detachable mountings, connections and couplings or integrated, and may be direct and via media indirect mountings, connections, and couplings, and also may be inner mountings, connections and couplings of two components or interaction relations between two components. For those skilled in the art, the specific meaning of the above-mentioned terms in the embodiments of the present disclosure can be understood according to specific circumstances.

[0103] In the present disclosure, unless specified or limited otherwise, the first feature is “on” or “under” the second feature refers to the first feature and the second feature may be direct or via media indirect contact. And, the first feature is “on,” “above,” “over” the second feature may refer to the first feature is right over the second feature or is diagonal above the second feature, or just refer to the horizontal height of the first feature is higher than the horizontal height of the second feature. The first feature is “below” or “under” the second feature may refer to the first feature is right below the second feature or is diagonal under the second feature, or just refer to the horizontal height of the first feature is lower than the horizontal height of the second feature.

[0104] Reference throughout this specification to “an embodiment,” “some embodiments,” “an example,” “a specific example,” or “some examples” means that a particular feature, structure, material, or characteristic described in connection with the embodiment or example is included in at least one embodiment or example of the present disclosure. The appearances of the above phrases in various places throughout this specification are not necessarily referring to the same embodiment or example of the present disclosure. Furthermore, the particular features, structures, materials, or characteristics may be combined in any suitable manner in one or more embodiments or examples. In addition, different embodiments or examples and features of different embodiments or examples described in the specification may be combined by those skilled in the art without mutual contradiction.

[0105] Although embodiments of the present disclosure have been illustrated and described, it is conceivable for those of ordinary skill in the art that various changes, modifications, replacements, and variations can be made to these embodiments without departing from the principles and spirit of the present disclosure. The scope of the present disclosure shall be defined by the claims as appended and their equivalents.

1-11. (canceled)

12. A window air conditioner comprising:

an indoor unit component including an indoor unit body;
and

an outdoor unit component including an outdoor unit body spaced apart from the indoor unit body in a longitudinal direction;

wherein the indoor unit component is rotatably connected to the outdoor unit component, to enable the outdoor unit component and the indoor unit component to rotate relative to each other about a rotation axis extending in a transverse direction.

13. The window air conditioner according to claim 12, wherein the outdoor unit body is reciprocally rotatable about the rotation axis between:

a first state, in which a back plate of the outdoor unit body is vertically arranged, and the rotation axis is located at a position at which a top portion of the back plate of the outdoor unit body is located; and

a second state, in which the back plate of the outdoor unit body is transversely arranged.

14. The window air conditioner according to claim 13, wherein:

the indoor unit component further includes a connection support configured to pass through a window opening, the connection support including an outer end extending to be pivotally connected to an upper inner end of the outdoor unit body, to enable the rotation axis to be located at an intersection between the back plate and a top plate of the outdoor unit body; and

the back plate of the outdoor unit body is flush with a bottom plate of the connection support when the outdoor unit body is in the second state.

15. The window air conditioner according to claim 13, wherein the back plate of the outdoor unit body includes a guide structure configured to be engaged with a guide rail at a mounting support mounted at the window opening to allow a drawable guiding of the outdoor unit body in an inward-outward direction in the second state.

16. The window air conditioner according to claim 15, wherein:

the guide structure is a first guide structure; and

the connection support includes a second guide structure at a bottom of the connection support, the second guide structure being configured to be engaged with the guide rail after the first guide structure is engaged with the guide rail.

17. The window air conditioner according to claim 13, further comprising:

a limit structure configured to prevent the outdoor unit body from further rotating in a current rotation direction in response to the outdoor unit body rotating into the second state in a process of rotating relative to the indoor unit component.

18. The window air conditioner according to claim 13, further comprising:

a limit structure configured to prevent, in response to the outdoor unit body rotating into the second state in a process of rotating relative to the indoor unit component, the outdoor unit body from rotating reversely in a direction along which the outdoor unit body returns to the first state.

19. The window air conditioner according to claim 18, wherein:

the limit structure includes:

a first buckle arranged at the indoor unit component;
and

a second buckle arranged at the outdoor unit component;

when the outdoor unit body is in the second state, the first buckle is engaged with the second buckle for locking to prevent the outdoor unit body from rotating reversely in the direction along which the outdoor unit body returns into the first state; and

the outdoor unit body is out of the second state when the first buckle is disengaged from the second buckle for unlocking.

20. The window air conditioner according to claim 18, wherein:

the limit structure includes:

a pull rod; and

a slide groove extending in the longitudinal direction and formed at the indoor unit component;

the slide groove has a locking groove at an inner end of the slide groove, the locking groove extending upwardly;

the pull rod includes a first end rotatably connected to the outdoor unit body and a second end engaged with the slide groove to be slidable along the slide groove; and the limit structure is configured to prevent, in response to the second end being engaged in the locking groove, the outdoor unit body from rotating reversely in the direction along which the outdoor unit body returns to the first state.

21. The window air conditioner according to claim **12**, wherein the indoor unit component is pivotally connected to the outdoor unit component.

22. The window air conditioner according to claim **12**, wherein the outdoor unit body has a grip groove located on at least one of a transverse side plate, a top plate, or a back plate of the outdoor unit body.

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